TEXTBOOK · FALL2 0 2 6

Linear Algebra I

Weeks 5–7

All questions and all formulas from each course file and assigned textbook sections — geometry and elimination, matrix operations and decompositions, transposes, permutations and vector spaces — each problem with a complete step-by-step solution. Compiled for independent study and Midterm preparation

Tashkent Technological University · Fall 2026

Sources: course notes, lectures, ISA, practical sessions; G. Strang, Introduction to Linear Algebra $(4e)$ and Linear Algebra and its Applications $(4e)$

QUESTIONBANK · FORMULACATALOG · FULLSOLUTIONSOCTOBER2 0 2 6

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1 Course Map and How to Use This Document

Week	Linear Algebra I
5	Geometry and elimination: row picture and column picture, Gaussian elimination, back substitution, elimination matrices E, permutation matrices P, connection with rank, $A = LU$ (Strang §§2.1–2.3)
6	Matrix operations: multiplication rules and four interpretations of AB, special matrices, inverses and their derivation, elimination matrices, LU/LDU decompositions, transpositions, permutations (Strang §§2.4–2.7)
7	Transpositions and permutations: transposition identities, symmetric matrices, six 3×3 permutations; vector spaces and subspaces, column space $C(A), null$ space $N(A)$ (Strang §§2.7, 3.1)

This guide continues the official collection "Linear Algebra I — Weeks 1–4" and covers weeks $5-7of$ the fall semester 2026. It collects every question and every formula from each course file and the assigned textbook sections — each problem with a complete step-by-step solution, as in the main collection.

Course files covered (weeks 5–7): lectures; practical sessions and in-class problems with solutions; Independent Study Assignments (ISA); the expanded week 5 Beamer theory notes; EXTRA material (week 6 review with examples). Textbooks: G. Strang, Introduction to Linear Algebra $(4thed.)$, with the parallel companion Linear Algebra and Its Applications $(4thed.)$.

How each week is organised. First comes the formula catalog: all definitions, theorems and rules of the week with explanations. Then, for every source file, a "questions and complete solutions" section: in-class problems, the practical session, Independent Study Assignments (ISA) and extra material. Problems are reproduced verbatim; solutions are complete, step by step, with checks.

Notation. Matrices are written $rowbyrow$: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ — first $rowab$, second $rowcd$; a column vector is written $\begin{bmatrix} x \\ y \end{bmatrix}$, $arow$ vector $\begin{bmatrix} x & y \end{bmatrix}$. Powers and indices are written with ^{} and _: A^{-1}, A^T, a_{11}. Weeks 5–7 close the first half of the course: elimination and matrix operations, the decompositions $A = LU = LDU = PA$, transposes, permutations and vector spaces. Together with the weeks 1–4 material they form the base assumed known for the week 8 Midterm.

2 Week 5 — Geometry and Elimination (Linear Algebra I)

[Source: Beamer lecture notes week 5 "Geometry and Elimination" (Linear Algebra $I, TT2026$): Strang, Introduction to Linear

Algebra, 4th ed., §§2.1–2.3; Linear Algebra and Its Applications, §§1.2–1.4]

2.6.1 Connection with rank: n pivots mean invertibility

Review: $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} has$rank$1and$ no inverse; $Ax = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$ has infinitely many solutions, while $Ax = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$ has none. Key fact: for an $n \times n$ square matrix, having n pivots is

equivalent to invertibility—then $Ax = bhasa$ unique solution for every b; the number of pivots found by elimination is the rank. Both algorithms act on $[A|b]$: Gauss–Jordan reduces the matrix to reduced echelon form; Gaussian elimination stops at upper triangular U and completes with back substitution.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} : \text{rank} A = 1, A^{-1} does not exist;$$

$$Ax = \begin{bmatrix} 3 \\ 6 \end{bmatrix} — infinitely many solutions,$$

$$Ax = \begin{bmatrix} 3 \\ 7 \end{bmatrix} — no solution$$

2.6.2 Row picture: rows as lines and planes

$$2x - y = 1, x + y = 5;$$

$$\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

Each $rowis$ one geometric constraint: in 2D lines, in 3D planes; the solution is their common point. The example lines intersect at $(x, y) = (2, 3)$; three planes in 3D:

$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix} \cdot \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix} — planes intersect at (u, v, w) = (1, 1, 2)$$

2.6.3 Column picture: $Axasa$ linear combination of columns

$$x \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} + y \cdot \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix};$$

$$check:$$

$$2 \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 3 \cdot \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

Strang emphasizes: $Axisa$ **linear combination of the columns of** A**—we** choose coefficients so that $x_1 \cdot a_1 + \ldots + x_n \cdot a_n = b$. Three-dimensional version (weights (1, 1, 2)):

$$u \cdot \begin{bmatrix} 2 \\ 4 \\ -2 \end{bmatrix} + v \cdot \begin{bmatrix} 1 \\ -6 \\ 7 \end{bmatrix} + w \cdot \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}$$

Both pictures are the same equation $Ax = b$: rows are intersections (equations), columns are linear combinations (vectors).

2.6.4 Three possible outcomes

- One solution—independent constraints. • Infinitely many solutions: $x + y = 2$ and 2x + 2y = 4—the second equation is

twice the first, the lines coincide.

- No solutions: $x + y = 2$ and 2x + 2y = 5—the left sides are proportional, the

right sides are not, the lines are parallel.

$$elimination reveals outcome : [00\dots0|0] \to 0 = 0—infinitely many solutions;$$
$$[00\dots0|c], c \neq 0,$$

$$\to 0 = c—no solutions$$

2.6.5 Pivot, multiplier, and forward elimination

$$\lambda_{ij} = \frac{a_{ij}}{a_{jj}}, operation R_i \leftarrow R_i - \lambda_{ij} \cdot R_j$$

To eliminate element a_{ij} below a nonzero pivot $a_{jj}(the$ multiplier is denoted λ_{ij} here), subtract $rowj$ multiplied by λ_{ij} from $rowi$. Running example: $\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 4 & -6 & 0 & -2 \\ -2 & 7 & 2 & 9 \end{array}\right]$; pivot 2, multipliers $\lambda_{21} = \frac{4}{2} = 2 and \lambda_{31} = -\frac{2}{2} = -1$:

$$R_2 \leftarrow R_2 - 2R_1, R_3 \leftarrow R_3 + R_1 :$$
$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 8 & 3 & 14 \end{array}\right]$$

Second pivot -8, multiplier $\lambda_{32} = 8/(-8) = -1$, operation $R_3 \leftarrow R_3 + R_2$. Triangular system and result of forward elimination:

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 0 & 1 & 2 \end{array}\right],$$
$$U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{bmatrix}, pivot elements 2,$$
$$-8, 1$$

2.6.6 Back substitution

$$w = 2; -8v - 2w = -12 \to v = 1; 2u + v + w = 5 \to u = 1;$$

$$x = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

The triangular system is solved from bottom up: the last equation contains only w, the second to last contains $vandw$. The answer is verified by substitution into the original equation: $\begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}$. $Arow$ operation is reversible (add 2R_1} back), so elimination preserves solutions; but the operation is applied to both left and right sides, otherwise the equations are not equivalent.

2.6.7 Elimination matrices E

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$E_{21} \cdot A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ -2 & 7 & 2 \end{bmatrix},$$

$$E_{21} \cdot b = \begin{bmatrix} 5 \\ -12 \\ 9 \end{bmatrix}$$

Left multiplication performs $arow$ operation, and the right side is transformed by the same matrix: $Ax = b$ becomes $(E_{21}A)x = E_{21}b$. Full pass to $U(the$ rightmost multiplier acts first—order of multiplication matters):

$$E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix},$$

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix},$$

$$E_{32} \cdot E_{31} \cdot E_{21} \cdot A = U$$

Undo step: subtraction is replaced by addition—for $E = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$ the inverse is $E^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} andE^{-1} \cdot E = I.$

2.6.8 Zero pivot: permutation matrices P

$$\left[\begin{array}{cc|c} 0 & 2 & 4 \\ 3 & 1 & 7 \end{array}\right] \to \left[\begin{array}{cc|c} 3 & 1 & 7 \\ 0 & 2 & 4 \end{array}\right] (rowexchange),$$

$$theny = 2, x = \frac{5}{3}$$

A zero pivot does not necessarily mean singularity: if there is a nonzero element below, rows are swapped. The swap is performed by a permutation matrix, and it also transforms the right side:

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

$$P \cdot \begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix},$$

$$P \cdot \begin{bmatrix} 4 \\ 7 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \end{bmatrix},$$

$$equivalentsystemPAx = Pb$$

Properties of P: one 1 in each row and column, rest zeros; $P^{-1} = P^T$; for a single swap $P^2 = I$; swapping rows $1and3$ — $P_{13} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$; left multiplication changes equations, not unknowns. Elimination failures: temporary — zero pivot with nonzero element below (swap saves); permanent — no available pivot in column (singularity: no solution or infinitely many).

2.6.9 Looking ahead: $A = LUandPA = LU$

$$withoutrowexchangesA = LU(U—uppertriangular,$$
$$Lstoreseliminationfactors); withexchanges$$

$$PA = LU$$

Week summary: rows — intersections, columns — combinations; elimination — zeros below pivots plus back substitution; E — row replacement, P — row swap. LU decomposition $(A = LU, PA = LU)$ is built in week 6.

[Source: Week 5 Class Problems (Projection), problems $1 - 15(law5$classg$01 - g11$ and text layer pp. 23–31)]

Exercise 1. Row picture and column picture. Solve the system $\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$ **and describe both geometric pictures.**

Solution. Add equations: 3x $= 6, x = 2$, then $y = 5 - 2 = 3$. Lines 2x $-y = 1 and x + y = 5$ intersect at (2, 3); by columns: $2 \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 3 \cdot \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 & & 3 \\ 2 & + & 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$. Answer: $x = 2, y = 3$.

Exercise 2. Three outcomes. Classify the system $\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ c \end{bmatrix} for c = 4 and c = 5$**.**

Solution. $R_2 \leftarrow R_2 - 2R_1$ gives $row[00|c-4]$. When $c = 4$ this is $0 = 0$: $x + y = 2$ remains, infinitely many solutions — $\begin{bmatrix} 2 & t \\ t & \end{bmatrix}$. When $c = 5$ this is $0 = 1$ — contradiction, no solution.

Exercise 3. Find the multiplier. Eliminate below the first pivot in the augmented matrix $\left[\begin{array}{cc|c} 2 & 3 & 7 \\ 8 & 13 & 29 \end{array}\right]$ **and find both pivots.**

Solution. Multiplier $\lambda_{21} = \frac{8}{2} = 4$. $R_2 \leftarrow R_2 - 4R_1$: $[8, 13|29] - 4 \cdot [2, 3|7] = [0, 1|1]$. Result: $\left[\begin{array}{cc|c} 2 & 3 & 7 \\ 0 & 1 & 1 \end{array}\right]$; pivots are $2 and 1$.

Exercise 4. Back substitution. Solve the system $\begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ 0 & 0 & 4 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \\ 8 \end{bmatrix}$**.**

Solution. Bottom up: 4z $= 8, z = 2$; 3y $+2 \cdot 2 = 7, y = 1$; 2x $+1 - 2 = 3, x = 2$. Answer: $x = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$. Check: $2 \cdot 2 + 1 - 2 = 3$, $3 + 4 = 7$, $8 = 8$ — correct.

Exercise 5. Full elimination. Solve the system

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 1 & 10 \\ 1 & 2 & 3 & 14 \end{array}\right]$$

by elimination.

Solution. $R_2 \leftarrow R_2 - 2R_1 \to [01 - 1| - 2]$; $R_3 \leftarrow R_3 - R_1 \to [012|8]$; $R_3 \leftarrow R_3 - R_2 \to [003|10]$, so 3z $= 10$. Back substitution: $z = \frac{10}{3}$; $y - z = -2 \to y = \frac{4}{3}$; $x + y + z = 6 \to x = \frac{4}{3}$. Answer: [4/3; 4/3; 10/3]. Check: rows give 6, 10, 14 — correct.

Exercise 6. Identify elementary matrix. Find E **that performs** $R_3 \leftarrow R_3 - 4R_1 ona 3 \times 3$ **matrix.**

Solution. E is identity matrix with −4 in position (3, 1): $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$; $row 3 of EA$ equals $(row 3) - 4 \cdot (row 1)$, rows 1–2 unchanged. Inverse operation is addition: $E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}$ with $+4$ in position (3, 1).

Exercise 7. EA **multiplication. Let** $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} andA = \begin{bmatrix} 3 & 1 \\ 6 & 5 \end{bmatrix}$**. Compute** EA **and explain the operation performed.**

Solution. Row 1 of product: $\begin{bmatrix} 3 & 1 \end{bmatrix}$; $row2$: $-2 \cdot \begin{bmatrix} 3 & 1 \end{bmatrix} + \begin{bmatrix} 6 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 3 \end{bmatrix}$. $SoEA = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$: E subtracts twice $row1$ from $row2$ — elimination with multiplier $\frac{6}{3} = 2$.

Exercise 8. Transformed right side. For $A = \begin{bmatrix} 3 & 1 \\ 6 & 5 \end{bmatrix} andb = \begin{bmatrix} 4 \\ 17 \end{bmatrix} useE = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$ **to solve the system.**

Solution. $Eb = \begin{bmatrix} 4 \\ -8 \quad + \quad 17 \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \end{bmatrix}$; together with $EA = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ (exercise $7) we$ get 3y $= 9, y = 3$ and 3x $+3 = 4, x = \frac{1}{3}$. Check in original system: $3 \cdot (\frac{1}{3}) + 3 = 4$, $6 \cdot (\frac{1}{3}) + 5 \cdot 3 = 17$ — correct.

Exercise 9. Zero pivot. Solve the system $\begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix} byrow$ **swap; find** P**.**

Solution. Zero in position (1, 1) is temporary failure: $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $PA = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$, $Pb = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$. Then 2y $= 4, y = 2$; 3x $+y = 7, x = \frac{5}{3}$. Check: $2 \cdot 2 = 4, 3 \cdot (\frac{5}{3}) + 2 = 7$ — correct; zero pivot didn't mean singularity.

Exercise 10. Three-row **permutation. Find** P **that swaps rows** $1 and 3$ **of matrix** $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 4 & 0 & 2 \end{bmatrix}$**; compute** PA**.**

Solution. $P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ — rows of identity matrix in order 3, 2, 1; $row1 of PA$ equals $row3$ of matrix $A and so$ on: $PA = \begin{bmatrix} 4 & 0 & 2 \\ 0 & 1 & 3 \\ 1 & 2 & 0 \end{bmatrix}$. Swap $1 \leftrightarrow 3$ is involution: $P^{-1} = P and P^2 = I$ (consistent with $P^{-1} = P^T$).

Exercise 11. Singular but consistent. Solve the system $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$**.**

Solution. Row 2 with the right-hand side $(6 = 2 \cdot 3)$ is twice $row1$, so $R_2 \leftarrow R_2 - 2R_1$ gives $0 = 0$. We are left with $x+$ 2y $+z = 3 and y + z = 2$; free variable $z = t$: $y = 2 - t$, $x+2(2-t)+t = 3 \to x = t-1$. Answer: $\begin{bmatrix} t & 1 \\ 2 & t \\ t & \end{bmatrix}$ — infinitely many solutions. Verification

for $t = 1$: $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ gives 3, 6, 2 — correct.

Exercise 12. Degenerate and inconsistent. Replace the right-hand side in the previous system with $b = \begin{bmatrix} 3 \\ 7 \\ 2 \end{bmatrix}$. Is the system consistent?

Solution. Now $7 \neq 2 \cdot 3$. $R_2 \leftarrow R_2 - 2R_1$ gives the $row [000|7-6] = [000|1]$, that is, $0 = 1$ — a contradiction. The system is inconsistent: the right-hand side violates the necessary condition $b_2 = 2b_1$.

Exercise 13. Column interpretation of consistency. For $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, which vectors $b = [b_1; b_2]$ admit a solution $Ax = b$?

Solution. The second row is twice the first: after $R_2 \leftarrow R_2 - 2R_1$ we have $[00|b_2 - 2b_1]$, so consistency requires $b_2 = 2b_1$. The columns $\begin{bmatrix} 1 \\ 2 \end{bmatrix} and \begin{bmatrix} 2 \\ 4 \end{bmatrix} = 2 \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ are multiples of $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, so Ax always lies on the line $b_2 = 2b_1$. Answer: $b = [b_1;\ 2b_1]$ — b must lie on the line through the columns of A.

Exercise 14. Order of matrix operations. Let $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} andP = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Compare $EPandPE$.

Solution. EP: $row1 = \begin{bmatrix} 0 & 1 \end{bmatrix}$, $row2 = -2 \cdot \begin{bmatrix} 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \end{bmatrix} = [1, -2]$, $i.e.$ $EP = \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}$. PE: $row1 = 1 \cdot \begin{bmatrix} -2 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \end{bmatrix}$, $row2 = \begin{bmatrix} 1 & 0 \end{bmatrix}$, $i.e.$ $PE = \begin{bmatrix} -2 & 1 \\ 1 & 0 \end{bmatrix}$. The products are different: row operations, in general, do not commute.

Exercise 15. Challenge: construct a system. Build $a 3 \times 3$ system requiring row exchange, with three pivots and solution $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

Solution. A zero in position (1, 1) with a nonzero element below: $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$; set the right-hand side by the column picture: $b = A \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$. Swapping rows $1 and 2 (P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix})$ turns A into the identity matrix: pivots 1, 1, 1, solution $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

Verification: $A \cdot \begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}2\\1\\3\end{bmatrix}$ — correct.

[Source: Week 5 Practical Session problems with solutions (Week$5TA$ Problems Projection), problems 1–15; matrices restored
from the text layer]

Exercise 1. System of two lines. Solve the system $\begin{bmatrix}1 & 2\\3 & 1\end{bmatrix} \cdot \begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}7\\8\end{bmatrix}$.

Solution. $R_2 \leftarrow R_2 - 3R_1$: $(1-6)y = 8-21$, that is, $-5y = -13, y = \frac{13}{5}$; then $x = 7-2\cdot(\frac{13}{5}) = \frac{9}{5}$. Answer: (9/5, 13/5). Verification: $\frac{9}{5} + \frac{26}{5} = 7$, $\frac{27}{5} + \frac{13}{5} = 8$ — correct.

Exercise 2. Parallel or coincident? Classify the system $\begin{bmatrix}2 & 4\\1 & 2\end{bmatrix} \cdot \begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}6\\c\end{bmatrix} for c = 3 and c = 4$.

Solution. Row 1 is twice $row 2 on$ the left; consistency requires the same on the right: $6 = 2c$, that is, $c = 3$. For $c = 3$, we are left with one equation $x + 2y = 3$ — infinitely many solutions $\begin{bmatrix}3 & 2t\\t\end{bmatrix}$ (the lines coincide). For $c = 4$, doubling the second equation gives $2x + 4y = 8 \neq 6$ — no solutions (the lines are parallel).

Exercise 3. One elimination step. Eliminate below the first pivot in the matrix $\begin{bmatrix}3 & 2\\9 & 8\end{bmatrix}$.

Solution. The multiplier $\lambda_{21} = \frac{9}{3} = 3$; $R_2 \leftarrow R_2 - 3R_1$: $[9-9, 8-6] = [0, 2]$. Result: $\begin{bmatrix}3 & 2\\0 & 2\end{bmatrix}$; pivots $3 and 2$.

Exercise 4. Triangular system. Solve $\begin{bmatrix}1 & 2 & 1\\0 & 2 & -2\\0 & 0 & 3\end{bmatrix} \cdot \begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}8\\0\\6\end{bmatrix}$.

Solution. From bottom up: $3z = 6, z = 2$; $2y - 2\cdot 2 = 0, y = 2$; $x + 4 + 2 = 8, x = 2$. Answer: $\begin{bmatrix}2\\2\\2\end{bmatrix}$. Verification: $2 + 4 + 2 = 8$, $4 - 4 = 0$, $3 \cdot 2 = 6$ — correct.

Exercise 5. Complete system with three unknowns. Solve the system

$$\left[\begin{array}{ccc|c}1 & 1 & 0 & 3\\2 & 3 & 1 & 9\\1 & 0 & 1 & 4\end{array}\right]$$

.

Solution. $R_2 \leftarrow R_2 - 2R_1 \to [011|3]$; $R_3 \leftarrow R_3 - R_1 \to [0-11|1]$; $R_3 \leftarrow R_3 + R_2 \to [002|4]$, that is, $2z = 4, z = 2$. Backwards: $y + z = 3 \to y = 1$; $x + y = 3 \to x = 2$. Answer: $\begin{bmatrix}2\\1\\2\end{bmatrix}$.

Verification: $2 + 1 = 3$, $4 + 3 + 2 = 9$, $2 + 2 = 4$ — correct.

Exercise 6. Construct E_{32}. Write the 3×3 matrix for the operation $R_3 \leftarrow R_3+5R_2$.

Solution. $E_{32}is$ the identity matrix with 5 in position (3, 2): $E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix}$; rows 1–2 are unchanged, and $row3of$ the product equals $(row3) + 5 \cdot (row2)$.

Exercise 7. Reverse the operation. Find the inverse matrix for $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}$.

Solution. E implements $R_3 \leftarrow R_3 - 3R_2$ (subtracting three times $row2$); the inverse operation is addition: $E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$ (the sign changed). Verification: $E^{-1} \cdot E = I$.

Exercise 8. Transform the right-hand side. Apply $E = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $tob = \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}$.

Solution. $Eb = [3;\ 5 - 2 \cdot 3;\ 7] = \begin{bmatrix} 3 \\ -1 \\ 7 \end{bmatrix}$: the second component becomes $5 - 2 \cdot 3 = -1$. E eliminates x from the second equation with multiplier 2 — the right-hand side is transformed by the same operation.

Exercise 9. Row exchange matrix. Find P that swaps rows $2and3ofa3\times3$ matrix.

Solution. $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ is the identity matrix with rows $2and3$ swapped. A double swap returns the original order: $P^2 = I$, so $P^{-1} = P$ (consistent with $P^{-1} = P^T$); left multiplication swaps equations, not unknowns.

Exercise 10. System requiring exchange. Solve the system $\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 14 \end{bmatrix}$.

Solution. The pivot in position (1, 1) is zero — swap rows: 2x + 3y = $14andy = 4$. Then 2x $+12 = 14, x = 1$. Answer: (1, 4). Check: $0 \cdot 1 + 4 = 4, 2 + 12 = 14$ — correct.

Exercise 11. Free variable. Solve the system $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix}$.

Solution. The third $rowis0 = 0$, giving no constraints; there are two pivots and three unknowns. Let $z = t$: $y + t = 3 \to y = 3 - t$; $x + (3 - t) + t = 4 \to x = 1$. Answer: $\begin{bmatrix} 1 & \\ 3 & t \\ t & \end{bmatrix}$.

Check at $t = 0$: $\begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$ gives 4, 3, 0 — correct.

Exercise 12. Inconsistent row. Explain why the system

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 2 \\ 1 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

has **no solutions.**

Solution. The last row reads as $0 \cdot x + 0 \cdot y + 0 \cdot z = 1, i.e., \ 0 = 1$ — an impossible equality: a zero left-hand side cannot give a nonzero right-hand side for any x, y, z. The system is inconsistent.

Exercise 13. Compare $EPandPE$. For $E = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} andP = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ compute both products.

Solution. EP: $row1 = \begin{bmatrix} 0 & 1 \end{bmatrix}$, $row2 = 4 \cdot \begin{bmatrix} 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \end{bmatrix} = [1, 4]$, $i.e.$, $EP = \begin{bmatrix} 0 & 1 \\ 1 & 4 \end{bmatrix}$. PE: $row1 = 1 \cdot \begin{bmatrix} 4 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 1 \end{bmatrix}$, $row2 = \begin{bmatrix} 1 & 0 \end{bmatrix}$, $i.e.$, $PE = \begin{bmatrix} 4 & 1 \\ 1 & 0 \end{bmatrix}$. The products are different: ”first elimination, then exchange” and ”first exchange, then elimination” give different matrices.

Exercise 14. Column space test. Can the system $\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$ be solved?

Solution. Both columns have the second component twice the first: $\begin{bmatrix} 1 \\ 2 \end{bmatrix} and \begin{bmatrix} 3 \\ 6 \end{bmatrix} = 3 \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix}$; any combination $x \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \cdot \begin{bmatrix} 3 \\ 6 \end{bmatrix} has$ the form $\begin{bmatrix} s \\ 2s \end{bmatrix}$. For $b = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$ we have $5 \neq 2 \cdot 2 = 4$ — $bisnotin$ the column space. Answer: no (doubling the first equation gives 2x + 6y $= 4 \neq 5$ — the lines are parallel).

Exercise 15. Challenge: construct a permutation. Construct P that maps rows $(R_1, R_2, R_3)to(R_3, R_1, R_2)$.

Solution. New $row1isoldR_3$, new $2isoldR_1$, new $3isoldR_2$, so the rows of $Pare$ the rows of the identity matrix in order 3, 1, 2: $P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$. Check on rows $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$: $P \cdot A = \begin{bmatrix} 7 & 8 & 9 \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$. Left multiplication permutes rows; $P^3 = I$.

[Source: Week 5 Independent Study Assignments, practices 1–15; no solutions file provided — complete solutions with verification

are given below]

Practice 1. Row picture and column picture. Solve the system $\begin{bmatrix}2 & -1\\ 1 & 1\end{bmatrix}\cdot\begin{bmatrix}x\\ y\end{bmatrix}=\begin{bmatrix}1\\ 5\end{bmatrix}$**. Describe both geometric pictures. Show each elimination step and verify the answer.**

Solution. Add the equations: 3x $=6, x=2, y=3$. Check: $2\cdot 2-3=1, 2+3=5$. The lines 2x $-y=1 and x+y=5$ intersect at (2, 3); in columns: $2\cdot\begin{bmatrix}2\\1\end{bmatrix}+3\cdot\begin{bmatrix}-1\\1\end{bmatrix}=\begin{bmatrix}1\\5\end{bmatrix}$. Answer: $x=2, y=3$.

Practice 2. Three outcomes. Classify the system $\begin{bmatrix}1 & 1\\ 2 & 2\end{bmatrix}\cdot\begin{bmatrix}x\\ y\end{bmatrix}=\begin{bmatrix}2\\ c\end{bmatrix} for c=4 and c=5$**. Show each elimination step and verify the answer.**

Solution. $R_2\leftarrow R_2-2R_1$ gives $[00|c-4]$. For $c=4$: $0=0$, solutions $\begin{bmatrix}2 & t\\ t & \end{bmatrix}$ (check: $(2-t)+t=2, 2(2-t)+$ 2t $=4$). For $c=5$: $0=1$ — no solutions.

Practice 3. Find the multiplier. Eliminate below the first pivot in $\left[\begin{array}{cc|c}2 & 3 & 7\\ 8 & 13 & 29\end{array}\right]$**; find both pivots. Show each step and verify the answer.**

Solution. $\lambda_{21}=\frac{8}{2}=4$; $R_2\leftarrow R_2-4R_1$: $[8, 13|29]-4\cdot[2, 3|7]=[0, 1|1]$. Result: $\left[\begin{array}{cc|c}2 & 3 & 7\\ 0 & 1 & 1\end{array}\right]$, pivots are $2 and 1$. Check by invertibility: $[0, 1|1]+4\cdot[2, 3|7]=[8, 13|29]$.

Practice 4. Back substitution. Solve $\begin{bmatrix}2 & 1 & -1\\ 0 & 3 & 2\\ 0 & 0 & 4\end{bmatrix}\cdot\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}3\\ 7\\ 8\end{bmatrix}$**. Show each step and verify the answer.**

Solution. From bottom up: $z=\frac{8}{4}=2$; 3y $+2\cdot 2=7, y=1$; 2x $+1-2=3, x=2$. Verification: $\begin{bmatrix}2 & 1 & -1\\ 0 & 3 & 2\\ 0 & 0 & 4\end{bmatrix}\cdot\begin{bmatrix}2\\1\\2\end{bmatrix}=\begin{bmatrix}3\\7\\8\end{bmatrix}$. Answer: $x=\begin{bmatrix}2\\1\\2\end{bmatrix}$.

Practice 5. Complete elimination. Solve the system

$$\left[\begin{array}{ccc|c}1 & 1 & 1 & 6\\ 2 & 3 & 1 & 10\\ 1 & 2 & 3 & 14\end{array}\right]$$

by elimination. Show each step and verify the answer.

Solution. $R_2\leftarrow R_2-2R_1 and R_3\leftarrow R_3-R_1$ give $\left[\begin{array}{ccc|c}1 & 1 & 1 & 6\\ 0 & 1 & -1 & -2\\ 0 & 1 & 2 & 8\end{array}\right]$; then $R_3\leftarrow R_3-R_2$ gives [0 0 3 | 10]. Backwards: $z=\frac{10}{3}$, $y=z-2=\frac{4}{3}$, $x=6-y-z=\frac{4}{3}$. Verification: the rows give 6, 10, 14 — correct. Answer: [4/3; 4/3; 10/3].

Practice 6. Identify the elementary matrix. Find E that implements $R_3 \leftarrow R_3 - 4R_1 on a 3\times 3$ matrix. Show each step and verify the answer.

Solution. $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$ — the identity with -4 in position (3, 1); the third $row of EA is (row 3) - 4 \cdot (row 1)$. Verification on $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{bmatrix}$: $row 3 is \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} - 4 \cdot \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} -3 & -2 & -1 \end{bmatrix}$. Inverse: $E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}$.

Practice 7. Multiplication EA. Let $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$, $A = \begin{bmatrix} 3 & 1 \\ 6 & 5 \end{bmatrix}$. Compute EA and explain the operation. Show each step and verify the answer.

Solution. Row 1: $\begin{bmatrix} 3 & 1 \end{bmatrix}$; $row 2$: $\begin{bmatrix} 6 & 5 \end{bmatrix} - 2 \cdot \begin{bmatrix} 3 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 3 \end{bmatrix}$; $EA = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ — subtracting twice $row 1$ from $row 2$ (multiplier 2). Verification: $E^{-1} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ returns A.

Practice 8. Transformed right-hand side. For $A = \begin{bmatrix} 3 & 1 \\ 6 & 5 \end{bmatrix} and b = \begin{bmatrix} 4 \\ 17 \end{bmatrix} use E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$ to solve the system. Show each step and verify the answer.

Solution. $Eb = [4;\ -2 \cdot 4 + 17] = \begin{bmatrix} 4 \\ 9 \end{bmatrix}$; $EA = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$; then 3y $= 9 \rightarrow y = 3$ and 3x $+3 = 4 \rightarrow x = \frac{1}{3}$. Verification in the original system: $3 \cdot (\frac{1}{3}) + 3 = 4$, $6 \cdot (\frac{1}{3}) + 5 \cdot 3 = 17$ — correct.

Practice 9. Zero pivot element. Solve $\begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix} by row$ exchange; find P. Show each step and verify the answer.

Solution. $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$; $PA = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$, $Pb = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$; 2y $= 4 \rightarrow y = 2$, 3x $+2 = 7 \rightarrow x = \frac{5}{3}$. Verification: $0 \cdot (\frac{5}{3}) + 2 \cdot 2 = 4$, $3 \cdot (\frac{5}{3}) + 2 = 7$ — correct.

Practice 10. Permutation of three rows. Find P that swaps rows $1 and 3$ of matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 4 & 0 & 2 \end{bmatrix}$; compute PA. Show each step and verify the answer.

Solution. $P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$; $row 1 of PA = row 3 of A and row 3 of PA = row 1 of A$, so $PA =$

$\begin{bmatrix} 4 & 0 & 2 \\ 0 & 1 & 3 \\ 1 & 2 & 0 \end{bmatrix}$. Verification: $P^2 = I, i.e.,\ P \cdot PA = A;\ P^{-1} = P = P^T$.

Practice 11. Degenerate but consistent. Solve $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$**. Show each step and verify the answer.**

Solution. $R_2 \leftarrow R_2 - 2R_1$ gives $[000|6 - 2 \cdot 3] = [000|0]$ — $a row$ disappeared, the system is consistent but degenerate. Remaining: $x+$ 2y $+z = 3 and y + z = 2$; $z = t$: $y = 2 - t, x = t - 1$. Answer: $\begin{bmatrix} t & 1 \\ 2 & t \\ t & \end{bmatrix}$. Verification at $t = 2$: $\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$ gives 3, 6, 2 — correct.

Practice 12. Degenerate and inconsistent. In **the previous system, replace the right-hand side with** $b = \begin{bmatrix} 3 \\ 7 \\ 2 \end{bmatrix}$**.** Is **the system consistent? Show each step and verify the answer.**

Solution. $R_2 \leftarrow R_2 - 2R_1$ gives $[000|7 - 2 \cdot 3] = [000|1]$, $i.e.,\ 0 = 1$ — contradiction. The system is inconsistent: the necessary condition $b_2 = 2b_1$ is violated $(7 \neq 6)$.

Practice 13. Column interpretation of consistency. For $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$**, which vectors** $b = [b_1;\ b_2]$ **admit** a **solution to** $Ax = b$**? Show each step and verify the answer.**

Solution. $R_2 \leftarrow R_2 - 2R_1$ leaves $[12|b_1;\ 00|b_2 - 2b_1]$: consistency requires $b_2 = 2b_1$. The columns $\begin{bmatrix} 1 \\ 2 \end{bmatrix} and \begin{bmatrix} 2 \\ 4 \end{bmatrix} = 2 \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ are multiples of $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, so Ax always lies on the line $b_2 = 2b_1$. Answer: $b = [b_1;\ 2b_1]$. Verification: $\begin{bmatrix} 4 \\ 8 \end{bmatrix}$ is solved $(x = 4,\ y = 0)$, $\begin{bmatrix} 4 \\ 9 \end{bmatrix} is$ not.

Practice 14. Order of matrix operations. Let $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} and P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$**. Compare** $EP and PE$**. Show each step and verify the answer.**

Solution. EP: rows $\begin{bmatrix} 0 & 1 \end{bmatrix} and [1,\ -2]$, $i.e.,\ EP = \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}$. PE: rows $\begin{bmatrix} -2 & 1 \end{bmatrix} and \begin{bmatrix} 1 & 0 \end{bmatrix}$, $i.e.,\ PE = \begin{bmatrix} -2 & 1 \\ 1 & 0 \end{bmatrix}$. The products are different — row operations do not commute: in EP elimination acts on the permuted rows, in $PE on$ the original ones.

Practice 15. Challenge: construct a **system. Build** $a 3 \times 3$ **system that requires** row **exchange, with three pivots and solution** $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$**. Show each step and verify the answer.**

Solution. $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ — zero in position (1, 1), with a one below; $b = A \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$. Swapping rows $1 and 2 (P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix})$ gives the identity matrix: pivots 1, 1, 1, solution $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Verification: $A \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$; without exchange, the first column is not eliminated.

3 Week 6 — Matrix Operations, Inverses and Decompositions (Linear Algebra I)

[Source: Week 6 Lecture presentation (Strang, Introduction to Linear Algebra, 4th ed., §§2.4–2.7; companion: Lay, §§1.4–1.6)]

3.5.1 Review of week 5: two pictures of the system and elimination

Row picture: equations are intersecting lines or planes. Column picture: $Ax = x_1 \cdot a_1 + ... + x_n \cdot a_n = b$. Example: $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$ gives $(x, y) = (2, 1)$; both pictures describe the same system. Elimination: with pivot $a_{kk} \neq 0$, the multiplier $m_{ik} = a_{ik}/a_{kk}$ is taken and $R_i \leftarrow R_i - m_{ik} \cdot R_k$ is performed; thus $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 3 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, then back substitution solves the triangular system. Degeneracy: $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$ has infinitely many solutions, while changing $6 to 7$ gives the contradiction $0 = 1$. For square matrices, a full set of pivots means a unique solution for any right-hand side.

3.5.2 Multiplication rules and four interpretations of AB

$$A(\text{m×n}), B(\text{n×p}) \rightarrow AB(\text{m×p}), (AB)_{ij} = \sum_k a_{ik} \cdot b_{kj}$$

Example: $\begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 10 & 4 \end{bmatrix}$. Four views of AB: (1) element-wise — $row i$ of matrix A is multiplied by column j of matrix B; (2) column-wise — $AB = [Ab_1 ... Ab_p]$; (3) row-wise — each $row of AB is a$ combination of rows of B; (4) outer products — $AB = \sum_k a_k \cdot r_k$ (column k of matrix A times $row k$ of matrix B).

$$(AB)C = A(BC), A(B + C) = AB + AC, (A + B)C = AC + BC, AI = IA = A$$

$$generally speaking AB \neq BA; (A + B)^2 = A^2 + AB + BA + B^2 \neq A^2 + 2AB + B^2$$

An explicit example of non-commutativity: $A = \begin{bmatrix}1 & 1\\0 & 1\end{bmatrix}$, $B = \begin{bmatrix}1 & 0\\1 & 1\end{bmatrix}$ give $AB = \begin{bmatrix}2 & 1\\1 & 1\end{bmatrix}$, but $BA = \begin{bmatrix}1 & 1\\1 & 2\end{bmatrix}$ — horizontal and vertical shears depend on the order of application.

3.5.3 Special matrices and elimination matrices

The identity matrix I leaves vectors unchanged; diagonal $D = diag(d_1, ..., d_n)$ scales coordinates. Upper triangular matrices have zeros below the diagonal, lower triangular matrices have zeros above it; triangular systems are solved by forward/backward substitution. The elimination operation is itself a matrix: $R_2 \leftarrow R_2 - 3R_1$ corresponds to $E = \begin{bmatrix}1 & 0\\-3 & 1\end{bmatrix}$, and $E \cdot \begin{bmatrix}2 & 1\\6 & 5\end{bmatrix} = \begin{bmatrix}2 & 1\\0 & 2\end{bmatrix}$; the transformed system is EAx $= Eb$, and $E^{-1} = \begin{bmatrix}1 & 0\\3 & 1\end{bmatrix}$ restores the original rows.

3.5.4 The inverse matrix: definition, criteria, rules

$$AA^{-1} = A^{-1}A = I \rightarrow Ax = bhassolutionx = A^{-1}bforanyb$$

The inverse reverses the linear transformation. $IfAx = 0$ for nonzero x, the inverse does not exist. For a square A, the following are equivalent: the inverse exists $\Leftrightarrow$ all pivots are nonzero (after possible exchanges) $\Leftrightarrow$ columns are independent $\Leftrightarrow Ax = 0$ only when $x = 0 \Leftrightarrow Ax = bhasa$ unique solution for any b.

$$(AB)^{-1} = B^{-1}A^{-1}, (A^T)^{-1} = (A^{-1})^T, (A^{-1})^{-1} = A,$$
$$(A^k)^{-1} = (A^{-1})^k$$

The order of factors in the inverse of a product is reversed: $(AB)(B^{-1}A^{-1}) = I$.

3.5.5 The inverse of $2 \times 2and$ the Gauss-Jordan method

$$\begin{bmatrix}a & b\\c & d\end{bmatrix} \cdot \begin{bmatrix}d & b\\c & a\end{bmatrix} = (ad - bc) \cdot I \rightarrow$$
$$\begin{bmatrix}a & b\\c & d\end{bmatrix}^{-1} = (1/(ad - bc)) \cdot \begin{bmatrix}d & b\\c & a\end{bmatrix} whenad - bc \neq 0$$

Example: $\begin{bmatrix}2 & 1\\5 & 3\end{bmatrix}^{-1} = \begin{bmatrix}3 & -1\\-5 & 2\end{bmatrix}$, since det $= 1$. Gauss-Jordan method: $[A|I] \rightarrow [I|A^{-1}]$ using the same row operations on both halves: $\begin{bmatrix}1 & 2 & 1 & 0\\3 & 7 & 0 & 1\end{bmatrix} \rightarrow \begin{bmatrix}1 & 2 & 1 & 0\\0 & 1 & -3 & 1\end{bmatrix} \rightarrow \begin{bmatrix}1 & 0 & 7 & -2\\0 & 1 & -3 & 1\end{bmatrix}$, that is $\begin{bmatrix}1 & 2\\3 & 7\end{bmatrix}^{-1} = \begin{bmatrix}7 & -2\\-3 & 1\end{bmatrix}$. The columns of the inverse: if $A^{-1} = [x_1...x_n]$, then $Ax_j = e_j$ — finding the inverse means solving n systems; for a single system $Ax = b$, triangular elimination is usually preferable to explicit computation of A^{-1}.

3.5.6 LU decomposition and solving systems through factors

$$A = LU : elimination factors—below diagonal of unit lower triangular L,$$
$$elimination result—upper triangular U$$

$$\begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} (verify by multiplication)$$

$$E_k \cdots E_2 E_1 A = U \ \rightarrow \ A = E_1^{-1} E_2^{-1} \cdots E_k^{-1} U = LU$$

The inverse of ”subtract a multiple of $a row$” is ”add that multiple back”; the order of factors is crucial (the right operation acts first). 3×3 example: $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 5 & 4 \\ -2 & 2 & 3 \end{bmatrix}$ with multipliers $m_{21} = 2$, $m_{31} = -1$, $m_{32} = 1$ gives $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix} and U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & 2 \end{bmatrix}$ (verify $LU = A row by row$). Solving the system: first $Ly = b$ by forward substitution, then $Ux = y$ by backward substitution; for $L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 5 \\ 19 \end{bmatrix} we get y = \begin{bmatrix} 5 \\ 4 \end{bmatrix} and x = [\frac{3}{2}$; 2]. By decomposing A once, we can solve systems with many right-hand sides.

3.5.7 LDU decomposition, pivots and determinant

$$U = D \cdot (D^{-1}U), D contains reference elements of U;$$
$$A = L \cdot D \cdot (D^{-1}U) with unit diagonals for L and$$

$$D^{-1}U$$

$$\begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \cdot [1 \frac{1}{2};$$
$$01]$$

Pivots and determinant: for $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with $a \neq 0$, the pivots are $a and d - cb/a$, and their product is ad $-bc =$ det A. Without row exchanges, the determinant equals the product of pivots; the normalized LDU is unique when it exists with nonzero pivots.

3.5.8 Permutations, $PA = LU$ and partial pivoting

Why exchange rows: $A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$ is invertible, but its first pivot is zero; exchanging rows gives $PA = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$ — a zero pivot may indicate a poor row order, not singularity.

Permutation matrices permute rows of I; there is exactly one 1 in each row and each column. $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ exchanges rows: PA — rows of A, AP — columns of A; there are $n!$ permutations of n rows. Example: $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ maps $(x_1,\, x_2,\, x_3)^T to (x_2,\, x_3,\, x_1)^T$ and preserves vector lengths and scalar products.

$$P^{-1} = P^T (since P^T \cdot P = I); with exchanges PA = LU$$

Example: $A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}, P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, L = I, U = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$; with exchanges, it's important to correctly track the multipliers in L. Partial pivoting: $[10^{-8}1;\ 1\ 1]$ is nonsingular, but the tiny first pivot gives a multiplier of 10^8 — exchanging rows avoids this; the rule is to take the largest in magnitude element of the active column (below and including the pivot).

3.5.9 Transposition, symmetric matrices, LDL^T

$$(A^T)_{ij} = a_{ji}, (A+B)^T = A^T + B^T, (AB)^T = B^T \cdot A^T$$

Example: $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$. A symmetric matrix satisfies $A^T = A$. $If a$ symmetric $A has an LDU$ decomposition without exchanges, its normalized upper factor equals L^T: $A = LDL^T$; for example $\begin{bmatrix} 2 & 2 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

3.5.10 Applications: differences, sums, tridiagonal systems

The difference matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$ gives $Ax = (x_1,\, x_2 - x_1,\, x_3 - x_2)^T$; its inverse $A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ gives cumulative sums — the discrete analog of differentiation and integration. The tridiagonal $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ describes nearest neighbor connections (finite difference equations): its factors preserve the narrow band, and solving via LU uses the structure, avoiding the dense inverse matrix.

3.5.11 Week's Summary

- Matrix multiplication has several equivalent interpretations, but is

non-commutative.

- The inverse exists only for non-singular square matrices. • Elimination is encoded in $A = LU = LDU$. • Row exchanges are encoded by the matrix P: $PA = LU$. • Transpositions and permutations prepare for the vector spaces of week 7.

[Source: Week 6 Problems in class (Solved) — 20 problems with solutions, from 2×2 to advanced decompositions]

Exercise 1. 2×2 **inverse. Find** $A^{-1} for A = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$**. Verify by multiplication.**

Solution. $\det A = 2 \cdot 3 - 1 \cdot 2 = 4 \neq 0$. $Bythe 2 \times 2$ formula: $A^{-1} = (\frac{1}{4}) \cdot \begin{bmatrix} 3 & -1 \\ -2 & 2 \end{bmatrix} = [\frac{3}{4} - \frac{1}{4}$; $-1/2$ 1/2]. Verification: $A \cdot A^{-1} = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix} \cdot [\frac{3}{4} - \frac{1}{4}; -\frac{1}{2}\frac{1}{2}] = [\frac{3}{2} - \frac{1}{2}, -\frac{1}{2} + \frac{1}{2}; \frac{3}{2} - \frac{3}{2}, -\frac{1}{2} + \frac{3}{2}] = I$.

Exercise 2. 2×2 **system solution. Solve** $Ax = b$**, where** $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}, b = \begin{bmatrix} 8 \\ 6 \end{bmatrix}$**.**

Solution. System: 3x + 2y $= 8, x+$ 4y $= 6$. Multiplying the second equation by 3 and subtracting the first, we get 10y $= 10$, that is, $y = 1, x = 6 - 4 = 2$. Thus, $x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Verification: $A \cdot x = [3 \cdot 2 + 2; 2 + 4] = \begin{bmatrix} 8 \\ 6 \end{bmatrix} = b$.

Exercise 3. 2×2 **inverse. Find** $A^{-1} for A = \begin{bmatrix} 4 & 1 \\ 2 & 5 \end{bmatrix}$**. Verify by multiplication.**

Solution. $\det A = 4 \cdot 5 - 1 \cdot 2 = 18 \neq 0$. $A^{-1} = (\frac{1}{18}) \cdot \begin{bmatrix} 5 & -1 \\ -2 & 4 \end{bmatrix} = [\frac{5}{18} - \frac{1}{18}$; $-1/9$ 2/9]. Verification: $A \cdot A^{-1} = I$ — for example, element (1,1) equals $4 \cdot (\frac{5}{18}) - \frac{1}{9} = \frac{20}{18} - \frac{2}{18} = 1$, element (2,2) equals $-\frac{2}{18} + \frac{20}{18} = 1$, off-diagonal elements are zeros.

Exercise 4. Solving $a 2 \times 2$ **system. Solve** $Ax = b$**, where** $A = \begin{bmatrix} 2 & 2 \\ 1 & 6 \end{bmatrix}, b = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$**.**

Solution. System: 2x + 2y $= 4$ (that is, $x + y = 2), x+$ 6y $= 7$. Subtracting: 5y $= 5, y = 1, x = 1$. Thus, $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 2 & + & 2 \\ 1 & + & 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix} = b$.

Exercise 5. 2×2 **decomposition. Find** $A = LU = LDU for A = \begin{bmatrix} 2 & 1 \\ 2 & 4 \end{bmatrix}$**.**

Solution. Pivot element 2, multiplier $m_{21} = \frac{2}{2} = 1$: $R_2 \leftarrow R_2 - R_1$ gives $U = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$. We write the multiplier in L: $L = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $A = LU$ (verification: $\begin{bmatrix} 2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 4 \end{bmatrix}$). We take the pivot elements $2 and 3 to D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$; normalizing rows of U by dividing by pivot elements, we get $D^{-1}U = [1\frac{1}{2}$; 0 1]. Thus, $A = LU = L \cdot D \cdot D^{-1}U$.

Exercise 6. 2×2 **decomposition. Find** $A = LU = LDU for A = \begin{bmatrix} 3 & 1 \\ 3 & 5 \end{bmatrix}$.

Solution. $m_{21} = \frac{3}{3} = 1$: $R_2 \leftarrow R_2 - R_1$ gives $U = \begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix}$. $L = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $A = LU$ (verification: $\begin{bmatrix} 3 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 5 \end{bmatrix}$). $D = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$, $D^{-1}U = [1\frac{1}{3}; 0\ 1]$, and $A = L \cdot D \cdot D^{-1}U$.

Exercise 7. 2×2 **decomposition. Find** $A = LU = LDU for A = \begin{bmatrix} 1 & 1 \\ 4 & 3 \end{bmatrix}$.

Solution. $m_{21} = 4$: $R_2 \leftarrow R_2 - 4R_1$ gives $U = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$. $L = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$, $A = LU$ (verification: $4 \cdot \begin{bmatrix} 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -1 \end{bmatrix} = \begin{bmatrix} 4 & 3 \end{bmatrix}$). The second pivot element is negative — this is acceptable: $D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $D^{-1}U = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, and $A = L \cdot D \cdot D^{-1}U$.

Exercise 8. 2×2 **decomposition. Find** $A = LU = LDU for A = \begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix}$.

Solution. $m_{21} = \frac{5}{2}$: $R_2 \leftarrow R_2 - (\frac{5}{2})R_1$ gives $U = [21;\ 0\frac{3}{2}](4 - \frac{5}{2} = \frac{3}{2})$. $L = [10;\ \frac{5}{2}1], A = LU$ (verification: $(\frac{5}{2}) \cdot \begin{bmatrix} 2 & 1 \end{bmatrix} + [0\frac{3}{2}] = \begin{bmatrix} 5 & 4 \end{bmatrix}$). $D = [20;\ 0\frac{3}{2}], D^{-1}U = [1\frac{1}{2};\ 0\ 1]$, and $A = L \cdot D \cdot D^{-1}U$.

Exercise 9. 3×3 **elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 10 \\ 6 \end{bmatrix}$.

Solution. Augmented matrix $\begin{bmatrix} 1 & 3 & 1 & 6 \\ 2 & 5 & 2 & 10 \\ 1 & 4 & 3 & 6 \end{bmatrix}$. Step $1 (m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, -1, 0| - 2)$; $R_3 - R_1 = (0, 1, 2|0)$. Step $2 (m_{32} = -1)$: $R_3 + R_2 = (0, 0, 2| - 2)$. Triangular system: $x + 3y + z = 6, -y = -2, 2z = -2$. Back substitution: $z = -1, y = 2, x = 6 - 6 + 1 = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 1 & + & 6 & 1 \\ 2 & + & 10 & 2 \\ 1 & + & 8 & 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 6 \end{bmatrix} = b$.

Exercise 10. 3×3 **elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix}$.

Solution. The augmented matrix is $\begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 3 & 2 & 6 \\ 1 & 2 & 4 & 1 \end{bmatrix}$. Step $1 (m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, 1, 0|2)$; $R_3 - R_1 = (0, 1, 3| - 1)$. Step $2 (m_{32} = 1)$: $R_3 - R_2 = (0, 0, 3| - 3)$. Triangular system: $x + y + z = 2, y = 2, 3z = -3$. Back substitution: $z = -1, y = 2, x = 2 - 2 + 1 = 1$.

Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 1 & + & 2 & 1 \\ 2 & + & 6 & 2 \\ 1 & + & 4 & 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix} = b$.

Exercise 11. 3×3 **Elimination (degenerate case). Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 3 & 5 \end{bmatrix}$, $b = \begin{bmatrix} 4 \\ 8 \\ 2 \end{bmatrix}$**.**

Solution. The augmented matrix is $\begin{bmatrix} 1 & 2 & 1 & 4 \\ 2 & 4 & 2 & 8 \\ 1 & 3 & 5 & 2 \end{bmatrix}$. Step $1 (m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, 0, 0|0)$ — the second equation is exactly twice the first; $R_3 - R_1 = (0, 1, 4| - 2)$. After swapping rows $2 and 3$: $\begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & 1 & 4 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ — two pivots for three unknowns, one free variable. Let $z = t$: $y = -2 -$ 4t (from $y +$ 4z $= -2), x = 4 -$ 2y $- z = 8 +$ 7t. General solution $x = \begin{bmatrix} 8 & + & 7t \\ -2 & & 4t \\ t & & \end{bmatrix} = \begin{bmatrix} 8 \\ -2 \\ 0 \end{bmatrix} + t \cdot \begin{bmatrix} 7 \\ -4 \\ 1 \end{bmatrix}$, $t \in R$ — infinitely many solutions; the solution given in the file $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} isa$ particular solution at $t = -1$. Verification: $A \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 & + & 4 & 1 \\ 2 & + & 8 & 2 \\ 1 & + & 6 & 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \\ 2 \end{bmatrix} = b$. Geometrically: the first two equations define one plane, its intersection with the third is a line.

Exercise 12. 3×3 **Elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix}$**.**

Solution. The augmented matrix is $\begin{bmatrix} 1 & 3 & 1 & 6 \\ 2 & 5 & 2 & 10 \\ 1 & 4 & 4 & 5 \end{bmatrix}$. Step $1 (m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, -1, 0| - 2)$; $R_3 - R_1 = (0, 1, 3| - 1)$. Step $2 (m_{32} = -1)$: $R_3 + R_2 = (0, 0, 3| - 3)$. Triangular system: $x +$ 3y $+ z = 6, -y = -2,$ 3z $= -3$. Back substitution: $z = -1, y = 2, x = 6 - 6 + 1 = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 1 & + & 6 & 1 \\ 2 & + & 10 & 2 \\ 1 & + & 8 & 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix} = b$.

Exercise 13. $LU and LDU 3\times 3$**. Find** $LU and LDU for A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ -2 & 1 & 1 \end{bmatrix}$**.**

Solution. Multipliers $m_{21}=1, m_{31}=-1$: $R_2 - R_1 = (0,2,1)$; $R_3 + R_1 = (0,2,2)$. Second pivot is $2, m_{32}=1$: $R_3 - R_2 = (0,0,1)$. Thus, $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$; verification: $LU = A$ (third row: $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 1 \end{bmatrix}$). $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, normalized upper multiplier $D^{-1}U = [1\ \frac{1}{2}\ \frac{1}{2}$; 0 1 1/2; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Exercise 14. $LU and LDU 3\times 3$**. Find** $LU and LDU for A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 5 & 3 \\ -2 & 5 & 3 \end{bmatrix}$**.**

Solution. $m_{21}=2, m_{31}=-1$: $R_2 - 2R_1 = (0,3,1)$; $R_3 + R_1 = (0,6,4)$. Second pivot is $3, m_{32}=2$: $R_3 - 2R_2 = (0,0,2)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$; verification: third $row of LU is -\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + 2 \cdot \begin{bmatrix} 0 & 3 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 5 & 3 \end{bmatrix}$. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, $D^{-1}U = [1\ \frac{1}{2}\ \frac{1}{2}$; 0 1 1/3; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Exercise 15. $LU and LDU 3\times 3$**. Find** $LU and LDU for A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 5 & 2 \\ -2 & 11 & 5 \end{bmatrix}$**.**

Solution. $m_{21}=1, m_{31}=-1$: $R_2 - R_1 = (0,4,1)$; $R_3 + R_1 = (0,12,6)$. Second pivot is $4, m_{32}=3$: $R_3 - 3R_2 = (0,0,3)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 4 & 1 \\ 0 & 0 & 3 \end{bmatrix}$; verification: $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + 3 \cdot \begin{bmatrix} 0 & 4 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} -2 & 11 & 5 \end{bmatrix}$. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, $D^{-1}U = [1\ \frac{1}{2}\ \frac{1}{2}$; 0 1 1/4; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Exercise 16. $LU and LDU 3\times 3$**. Find** $LU and LDU for A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 4 & 3 \\ -2 & 1 & 1 \end{bmatrix}$**.**

Solution. $m_{21}=2, m_{31}=-1$: $R_2 - 2R_1 = (0,2,1)$; $R_3 + R_1 = (0,2,2)$. Second pivot is $2, m_{32}=1$: $R_3 - R_2 = (0,0,1)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$; verification: second

$row is 2 \cdot \begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 4 & 3 \end{bmatrix}$, third is $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 1 \end{bmatrix}$. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $D^{-1}U = [1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/2; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Exercise 17. Row Permutation and PLU. Find the row permutation P and factors satisfying $PA = LU$, for $A = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 3 & 2 \end{bmatrix}$.

Solution. The first diagonal element is zero — swap rows $1 and 2$: $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $PA = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 2 & 3 & 2 \end{bmatrix}$. Elimination: element (2, 1) is already zero ($m_{21} = 0$); $m_{31} = 2$: $R_3 - 2R_1 = (2, 3, 2) - 2 \cdot (1, 1, 0) = (0, 1, 2)$; then $m_{32} = \frac{1}{2}$: $R_3 - (\frac{1}{2})R_2 = (0, 1, 2) - (0, 1, \frac{1}{2}) = (0, 0, \frac{3}{2})$. Thus, $L = [100$; 0 1 0; $2\,\frac{1}{2}\,1], U = [110$; 0 2 1; 0 0 3/2]. Check: $LU = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 2 & 3 & 2 \end{bmatrix} = PA$ (third row: $2 \cdot (1, 1, 0) + (\frac{1}{2}) \cdot (0, 2, 1) + (0, 0, \frac{3}{2}) = (2, 3, 2)$). Thus, $PA = LU$: P reorders the equations, giving a suitable pivot; L contains the multipliers $2 and \frac{1}{2}$ with their original signs, although the operations subtract them.

Exercise 18. Transpose and inverse. For $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}$ compute A^T and verify $(A^T)^{-1} = (A^{-1})^T$.

Solution. Transpose: $A^T = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$. $\det A = 1 \cdot (1 \cdot 1 - 1 \cdot 0) - 2 \cdot (0 \cdot 1 - 1 \cdot 2) + 0 = 1 + 4 = 5$. By the adjugate matrix $(A^T)^{-1} = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix}$. Independently compute $A^{-1} = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{bmatrix}$; then $(A^{-1})^T = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix} = (A^T)^{-1}$. The equality holds; check: $A^T \cdot (A^T)^{-1} = I$.

Exercise 19. Symmetric LDL^T. Factor the symmetric $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix} as LDL^T$.

Solution. Pivot element 2; $m_{21} = \frac{2}{2} = 1 (m_{31} = 0)$: $R_2 - R_1 = (0, 3, 1)$. Second pivot element 3; $m_{32} = \frac{1}{3}$: $R_3 - (\frac{1}{3})R_2 = (0, 1, 3) - (0, 1, \frac{1}{3}) = (0, 0, \frac{8}{3})$. For symmetric A without

exchanges, the normalized upper multiplier is L^T, that is, $A = LDL^T$, where $L = [100; 1\ 1\ 0; 0\frac{1}{3}1], D = [200; 0\ 3\ 0; 0\ 0\ 8/3]$. Check: $L \cdot D = [200; 2\ 3\ 0; 0\ 1\ 8/3]$, and $(LD) \cdot L^T = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix} = A$.

Exercise 20. Challenge problem: tridiagonal model. Solve $Ax = b$ using $LU for A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$. Interpret the nearest neighbor structure.

Solution. Factorization: $m_{21} = -\frac{1}{2}$, so $R_2 \leftarrow R_2 + (\frac{1}{2})R_1$: $(-1,\ 2,\ -1) + (1,\ -\frac{1}{2},\ 0) = (0,\ \frac{3}{2},\ -1)$; $m_{32} = -\frac{2}{3}$, so $R_3 \leftarrow R_3 + (\frac{2}{3})R_2$: $(0,\ -1,\ 2) + (0,\ 1,\ -\frac{2}{3}) = (0,\ 0,\ \frac{4}{3})$. Thus, $L = [100; -1/2\ 1\ 0; 0 - \frac{2}{3}1], U = [2 - 10; 0\ 3/2\ -1; 0\ 0\ 4/3]$. Forward substitution $Ly = b$: $y_1 = 2$; $-(\frac{1}{2}) \cdot 2 + y_2 = 0$, so $y_2 = 1$; $-(\frac{2}{3}) \cdot 1 + y_3 = 2$, so $y_3 = \frac{8}{3}$. Back substitution $Ux = y$: $(\frac{4}{3})x_3 = \frac{8}{3}$, $x_3 = 2$; $(\frac{3}{2})x_2 - 2 = 1$, $x_2 = 2$; $2x_1 - 2 = 2, x_1 = 2$. Thus, $x = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$. Check: $A \cdot x = \begin{bmatrix} 4 & & 2 \\ -2 & + & 4 & 2 \\ -2 & + & 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = b$. The matrix is tridiagonal — nonzero elements only on the main diagonal and adjacent positions: each unknown is connected only to its nearest neighbors.

[Source: Week 6 Practical Session exercises with Solutions — 20 problems; Selected detailed solutions — detailed versions of
problems 9–12, 17, 20]

Exercise 1. 2×2 inverse. Find $A^{-1} for A = \begin{bmatrix} 3 & 2 \\ 1 & 6 \end{bmatrix}$. Verify by multiplication.

Solution. $\det A = 3 \cdot 6 - 2 \cdot 1 = 16 \neq 0$. $A^{-1} = (\frac{1}{16}) \cdot \begin{bmatrix} 6 & -2 \\ -1 & 3 \end{bmatrix} = [\frac{3}{8} - \frac{1}{8};\ -1/16\ 3/16]$. Check: $A \cdot A^{-1} = I$ — element $(1,1) is \frac{9}{8} - \frac{2}{16} = 1$, element $(2,2) is - \frac{1}{8} + \frac{18}{16} = 1$, off-diagonal elements are zero.

Exercise 2. 2×2 system solution. Solve $Ax = b$, where $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}, b = \begin{bmatrix} 14 \\ 12 \end{bmatrix}$.

Solution. System: 4x $+y = 14$, 2x + 3y $= 12$. Multiplying the first equation by 3 and subtracting the second, we get 10x $= 30$, that is, $x = 3, y = 14 - 12 = 2$. Thus, $x = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$. Check: $A \cdot x = \begin{bmatrix} 12 & + & 2 \\ 6 & + & 6 \end{bmatrix} = \begin{bmatrix} 14 \\ 12 \end{bmatrix} = b$.

Exercise 3. 2×2 inverse. Find $A^{-1} for A = \begin{bmatrix} 2 & 2 \\ 1 & 4 \end{bmatrix}$. Verify by multiplication.

Solution. $\det A = 2 \cdot 4 - 2 \cdot 1 = 6 \neq 0$. $A^{-1} = (\frac{1}{6}) \cdot \begin{bmatrix} 4 & -2 \\ -1 & 2 \end{bmatrix} = [\frac{2}{3} - \frac{1}{3};\ -1/6\ 1/3]$. Check:

$A \cdot A^{-1} = I$ — element $(1,1) is \frac{4}{3} - \frac{1}{3} = 1$, element $(2,2) is - \frac{1}{3} + \frac{4}{3} = 1$, off-diagonal elements are zero.

Exercise 4. 2×2 **system solution. Solve** $Ax = b$**, where** $A = \begin{bmatrix} 3 & 1 \\ 2 & 5 \end{bmatrix}, b = \begin{bmatrix} 8 \\ 14 \end{bmatrix}$**.**

Solution. System: 3x $+y = 8$, 2x + 5y $= 14$. Multiplying the first equation by 5 and subtracting the second, we get 13x $= 26$, that is, $x = 2, y = 8 - 6 = 2$. Thus, $x = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$.

Check: $A \cdot x = \begin{bmatrix} 6 & + & 2 \\ 4 & + & 10 \end{bmatrix} = \begin{bmatrix} 8 \\ 14 \end{bmatrix} = b$.

Exercise 5. 2×2 **factorization. Find** $A = LU = LDU for A = \begin{bmatrix} 3 & 1 \\ 5 & 5 \end{bmatrix}$**.**

Solution. $m_{21} = \frac{5}{3}$: $R_2 \leftarrow R_2 - (\frac{5}{3})R_1$ gives $U = [31;\ 0\frac{10}{3}] (5 - 5 = \frac{10}{3})$. $L = [10;\ \frac{5}{3}1], A = LU$ (check: $(\frac{5}{3}) \cdot \begin{bmatrix} 3 & 1 \end{bmatrix} + [0\frac{10}{3}] = \begin{bmatrix} 5 & 5 \end{bmatrix}$). $D = [30;\ 0\frac{10}{3}]$, $D^{-1}U = [1\frac{1}{3};\ 0\ 1]$, and $A = L \cdot D \cdot D^{-1}U$.

Exercise 6. 2×2 **factorization. Find** $A = LU = LDU for A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$**.**

Solution. $m_{21} = 2$: $R_2 \leftarrow R_2 - 2R_1$ gives $U = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. $L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $A = LU$ (check: $2 \cdot \begin{bmatrix} 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \end{bmatrix}$). Both pivots are 1, so $D = I and D^{-1}U = U$: $A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

Exercise 7. 2×2 **Factorization. Find** $A = LU = LDU for A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$**.**

Solution. $m_{21} = \frac{3}{2}$: $R_2 \leftarrow R_2 - (\frac{3}{2})R_1$ gives $U = [21;\ 0\frac{5}{2}] (4 - \frac{3}{2} = \frac{5}{2})$. $L = [10;\ \frac{3}{2}1], A = LU$ (check: $(\frac{3}{2}) \cdot \begin{bmatrix} 2 & 1 \end{bmatrix} + [0\frac{5}{2}] = \begin{bmatrix} 3 & 4 \end{bmatrix}$). $D = [20;\ 0\frac{5}{2}]$, $D^{-1}U = [1\frac{1}{2};\ 0\ 1]$, and $A = L \cdot D \cdot D^{-1}U$.

Exercise 8. 2×2 **Factorization. Find** $A = LU = LDU for A = \begin{bmatrix} 3 & 1 \\ 4 & 5 \end{bmatrix}$**.**

Solution. $m_{21} = \frac{4}{3}$: $R_2 \leftarrow R_2 - (\frac{4}{3})R_1$ gives $U = [31;\ 0\frac{11}{3}] (5 - \frac{4}{3} = \frac{11}{3})$. $L = [10;\ \frac{4}{3}1], A = LU$ (check: $(\frac{4}{3}) \cdot \begin{bmatrix} 3 & 1 \end{bmatrix} + [0\frac{11}{3}] = \begin{bmatrix} 4 & 5 \end{bmatrix}$). $D = [30;\ 0\frac{11}{3}]$, $D^{-1}U = [1\frac{1}{3};\ 0\ 1]$, and $A = L \cdot D \cdot D^{-1}U$.

Exercise 9. 3×3 **Elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 4 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix}$**.**

Solution. System: $x + y + z = 2$, 2x + 3y + 2z $= 6, x +$ 2y + 4z $= 1$; augmented matrix $\begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 3 & 2 & 6 \\ 1 & 2 & 4 & 1 \end{bmatrix}$. Eliminate $x (m_{21} = 2, m_{31} = 1)$: new second $row(2,\ 3,\ 2|6) - 2 \cdot (1,\ 1,\ 1|2) = (0,\ 1,\ 0|2)$; new third $row(1,\ 2,\ 4|1) - (1,\ 1,\ 1|2) = (0,\ 1,\ 3|-1)$. Eliminate $y (m_{32} = 1)$: (0,

$1,\ 3|-1)-(0,\ 1,\ 0|2)=(0,\ 0,\ 3|-3)$. Triangular system: $x+y+z=2, y=2,$ 3z $=-3$. Back substitution: $z=-1, y=2, x=2-2+1=1$. Thus, $x=\begin{bmatrix}1\\2\\-1\end{bmatrix}$. Check: $A\cdot x=$

$$\begin{bmatrix}1&+&2&1\\2&+&6&2\\1&+&4&4\end{bmatrix}=\begin{bmatrix}2\\6\\1\end{bmatrix}=b.$$

Exercise 10. 3×3 **Elimination (infinitely many solutions). Solve** $Ax=b$ **by Gaussian elimination for** $A=\begin{bmatrix}1&2&1\\2&4&2\\1&3&5\end{bmatrix}, b=\begin{bmatrix}4\\8\\2\end{bmatrix}$**.**

Solution. System: $x+$ 2y $+z=4,$ 2x + 4y + 2z $=8, x+$ 3y + 5z $=2$; augmented matrix $\begin{bmatrix}1&2&1&4\\2&4&2&8\\1&3&5&2\end{bmatrix}$. Eliminate $x(m_{21}=2, m_{31}=1)$: second row gives (0, 0, 0 | 0) — equation $0=0$ imposes no restrictions (second equation is exactly twice the first); third row: (0, 1, 4 | −2). Swapping rows $2 and 3$, we get echelon form $\begin{bmatrix}1&2&1&4\\0&1&4&-2\\0&0&0&0\end{bmatrix}$ — two pivots for three unknowns. Let $z=t, t\in R$: from second $row y=-2-$ 4t; from first $x=4-$ 2y $-z=8+$ 7t. All solutions: $x=\begin{bmatrix}8&+&7t\\-2&&4t\\t&&\end{bmatrix}=\begin{bmatrix}8\\-2\\0\end{bmatrix}+t\cdot\begin{bmatrix}7\\-4\\1\end{bmatrix}, t\in R.$

Check: $(8+7t)+2(-2-4t)+t=4 and (8+7t)+3(-2-4t)+$ 5t $=2$; second equation is automatically satisfied as twice the first. Geometrically: first two equations describe the same plane, and its intersection with the third is a line (solution family). Short answer file $x=\begin{bmatrix}1\\2\\-1\end{bmatrix}$ — particular solution at $t=-1$: $A\cdot\begin{bmatrix}1\\2\\-1\end{bmatrix}=\begin{bmatrix}4\\8\\2\end{bmatrix}=b.$

Exercise 11. 3×3 **Elimination. Solve** $Ax=b$ **by Gaussian elimination for** $A=\begin{bmatrix}1&3&1\\2&5&2\\1&4&4\end{bmatrix}, b=\begin{bmatrix}6\\10\\5\end{bmatrix}$**.**

Solution. System: $x+$ 3y $+z=6,$ 2x + 5y + 2z $=10, x+$ 4y + 4z $=5$; augmented matrix $\begin{bmatrix}1&3&1&6\\2&5&2&10\\1&4&4&5\end{bmatrix}$. Eliminate $x(m_{21}=2, m_{31}=1)$: second $row(0,-1,0|-2)$; third $row(0,1,3|-1)$. Second pivot −1; to eliminate the element below it, add $row 2 to row 3 (m_{32}=-1)$: $(0,\ 1,\ 3|-1)+(0,\ -1,\ 0|-2)=(0,\ 0,\ 3|-3)$. Triangular system: $x+$ 3y $+z=6, -y=-2,$ 3z $=-3$. Back substitution: $z=-1, y=2, x=6-6+1=1$. Thus, $x=\begin{bmatrix}1\\2\\-1\end{bmatrix}$. Check: $A\cdot x=$

$$\begin{bmatrix} 1 & + & 6 & 1 \\ 2 & + & 10 & 2 \\ 1 & + & 8 & 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix} = b.$$

Exercise 12. 3×3 **Elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix}$.

Solution. System: $x + y + z = 2$, 2x + 3y + 2z $= 6, x +$ 2y + 3z $= 2$; augmented matrix $\begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 3 & 2 & 6 \\ 1 & 2 & 3 & 2 \end{bmatrix}$. Eliminate $x(m_{21} = 2, m_{31} = 1)$: second $row(0, 1, 0|2)$; third $row(0, 1, 2|0)$. Eliminate $y(m_{32} = 1)$: $(0,\, 1,\, 2|0) - (0,\, 1,\, 0|2) = (0,\, 0,\, 2| - 2)$. Triangular system: $x + y + z = 2, y = 2$, 2z $= -2$. Back substitution: $z = -1, y = 2, x = 2 - 2 + 1 = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$.

Check: $A \cdot x = \begin{bmatrix} 1 & + & 2 & 1 \\ 2 & + & 6 & 2 \\ 1 & + & 4 & 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} = b.$

Exercise 13. $LUandLDU3 \times 3$**. Find** $LUandLDUforA = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 5 & 3 \\ -2 & 5 & 3 \end{bmatrix}$.

Solution. $m_{21} = 2, m_{31} = -1$: $R_2 - 2R_1 = (0, 3, 1)$; $R_3 + R_1 = (0, 6, 4)$. Second pivot element $3, m_{32} = 2$: $R_3 - 2R_2 = (0, 0, 2)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$; check: third $rowofLUis - \begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + 2 \cdot \begin{bmatrix} 0 & 3 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 5 & 3 \end{bmatrix}$. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, $D^{-1}U =$ [1 $\frac{1}{2}$ $\frac{1}{2}$; 0 1 1/3; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Exercise 14. $LUandLDU3 \times 3$**. Find** $LUandLDUforA = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 5 & 2 \\ -2 & 11 & 5 \end{bmatrix}$.

Solution. $m_{21} = 1, m_{31} = -1$: $R_2 - R_1 = (0, 4, 1)$; $R_3 + R_1 = (0, 12, 6)$. Second pivot element $4, m_{32} = 3$: $R_3 - 3R_2 = (0, 0, 3)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 4 & 1 \\ 0 & 0 & 3 \end{bmatrix}$; check: $- \begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + 3 \cdot \begin{bmatrix} 0 & 4 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} -2 & 11 & 5 \end{bmatrix}$. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, $D^{-1}U =$ [1 $\frac{1}{2}$ $\frac{1}{2}$; 0 1 1/4; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Exercise 15. $LUandLDU3\times3$**. Find** $LUandLDUforA=\begin{bmatrix}2&1&1\\4&4&3\\-2&1&1\end{bmatrix}$**.**

Solution. $m_{21}=2, m_{31}=-1$: $R_2-2R_1=(0,2,1)$; $R_3+R_1=(0,2,2)$. Second pivot element $2, m_{32}=1$: $R_3-R_2=(0,0,1)$. $L=\begin{bmatrix}1&0&0\\2&1&0\\-1&1&1\end{bmatrix}$, $U=\begin{bmatrix}2&1&1\\0&2&1\\0&0&1\end{bmatrix}$; check: $2\cdot\begin{bmatrix}2&1&1\end{bmatrix}+\begin{bmatrix}0&2&1\end{bmatrix}=\begin{bmatrix}4&4&3\end{bmatrix}and-\begin{bmatrix}2&1&1\end{bmatrix}+\begin{bmatrix}0&2&1\end{bmatrix}+\begin{bmatrix}0&0&1\end{bmatrix}=\begin{bmatrix}-2&1&1\end{bmatrix}$. $D=\begin{bmatrix}2&0&0\\0&2&0\\0&0&1\end{bmatrix}$, $D^{-1}U=[1\frac{1}{2}\frac{1}{2}$; 0 1 1/2; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Exercise 16. $LUandLDU3\times3$**. Find** $LUandLDUforA=\begin{bmatrix}2&1&1\\2&4&2\\-2&5&3\end{bmatrix}$**.**

Solution. $m_{21}=1, m_{31}=-1$: $R_2-R_1=(0,3,1)$; $R_3+R_1=(0,6,4)$. Second pivot element $3, m_{32}=2$: $R_3-2R_2=(0,0,2)$. $L=\begin{bmatrix}1&0&0\\1&1&0\\-1&2&1\end{bmatrix}$, $U=\begin{bmatrix}2&1&1\\0&3&1\\0&0&2\end{bmatrix}$; check: third $rowofLUis-\begin{bmatrix}2&1&1\end{bmatrix}+2\cdot\begin{bmatrix}0&3&1\end{bmatrix}+\begin{bmatrix}0&0&2\end{bmatrix}=\begin{bmatrix}-2&5&3\end{bmatrix}$. $D=\begin{bmatrix}2&0&0\\0&3&0\\0&0&2\end{bmatrix}$, $D^{-1}U=[1\frac{1}{2}\frac{1}{2}$; 0 1 1/3; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Exercise 17. Row Permutations and PLU**. Find the** row **permutation** P **and factors satisfying** $PA=LUforA=\begin{bmatrix}0&2&1\\1&1&0\\2&3&2\end{bmatrix}$**.**

Solution. (1) First diagonal element 0 — cannot use it as pivot; we swap rows $1and2$. The corresponding permutation matrix (same swap in I): $P=\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}$; multiplying $AbyPon$ the left swaps rows: $PA=\begin{bmatrix}1&1&0\\0&2&1\\2&3&2\end{bmatrix}$. (2) Eliminate below first pivot element 1: element (2, 1) is already zero ($m_{21}=0$); $m_{31}=\frac{2}{1}=2$, and $R_3\leftarrow R_3-2R_1$: $(2,3,2)-2\cdot(1,1,0)=(0,1,2)$. (3) Second pivot element 2; $m_{32}=\frac{1}{2}$, and $R_3\leftarrow R_3-(\frac{1}{2})R_2$: $(0,1,2)-(0,1,\frac{1}{2})=(0,0,\frac{3}{2})$; $U=[110$; 0 2 1; 0 0 3/2]. (4) Build L from multipliers: $L=[100$; $m_{21}10$; $m_{31}m_{32}1]=[100$; 0 1 0; 2 1/2 1]. (5) Check: $LU=\begin{bmatrix}1&1&0\\0&2&1\\2&3&2\end{bmatrix}=PA$ — third $rowis2\cdot(1,1,0)+(\frac{1}{2})\cdot(0,2,1)+(0,0,\frac{3}{2})=(2,3,2)$. Thus $PA=LU$: P reorders equations to provide a suitable pivot, and elimination completes the factorization; L contains the multipliers

$2and\frac{1}{2}$ with their original signs, although the operations subtract them.

Exercise 18. Transpose and Inverse. For $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}$ **compute** A^T **and verify** $(A^T)^{-1} = (A^{-1})^T$**.**

Solution. $A^T = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$. $\det A = 1 \cdot (1 \cdot 1 - 1 \cdot 0) - 2 \cdot (0 \cdot 1 - 1 \cdot 2) + 0 \cdot (0 \cdot 0 - 1 \cdot 2) =$ 5. From the adjugate matrix $(A^T)^{-1} = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix}$. Independently: $A^{-1} = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{bmatrix}$, from which $(A^{-1})^T = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix} = (A^T)^{-1}$. The equality holds; check: $A^T \cdot (A^T)^{-1} = I$.

Exercise 19. Symmetric LDL^T**. Factor the symmetric** $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix} asLDL^T$**.**

Solution. Pivot element 2; $m_{21} = 1 (m_{31} = 0)$: $R_2 - R_1 = (0, 3, 1)$. Second pivot element 3; $m_{32} = \frac{1}{3}$: $R_3 - (\frac{1}{3})R_2 = (0,\ 0,\ \frac{8}{3})$. For symmetric A without exchanges $A = LDL^T$, where $L = [100;\ 1\ 1\ 0;\ 0\frac{1}{3}1], D = [200;\ 0\ 3\ 0;\ 0\ 0\ 8/3]$. Check: $L \cdot D = [200;\ 2\ 3\ 0;\ 0\ 1\ 8/3]$, and $(LD) \cdot L^T = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix} = A$.

Exercise 20. Challenge problem: tridiagonal model. Solve $Ax = b$ **using** $LUforA = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, b = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$**. Interpret the nearest neighbor structure.**

Solution. (1) Decomposition: $m_{21} = (-1)/2 = -\frac{1}{2}$, so $R_2 \leftarrow R_2 + (\frac{1}{2})R_1$: $(-1,\ 2,\ -1) + (1,\ -\frac{1}{2},\ 0) = (0,\ \frac{3}{2},\ -1)$; $m_{32} = (-1)/(\frac{3}{2}) = -\frac{2}{3}$, so $R_3 \leftarrow R_3 + (\frac{2}{3})R_2$: $(0, -1, 2) + (0, 1, -\frac{2}{3}) = (0,\ 0,\ \frac{4}{3})$. Thus, $L = [100;\ -1/2\ 1\ 0;\ 0 - \frac{2}{3}1], U = [2 - 10;\ 0\ 3/2\ -1;\ 0\ 0\ 4/3]$; the multipliers appear in L with their original signs, including minuses. (2) Forward substitution $Ly = b$: $y_1 = 3$; $-(\frac{1}{2}) \cdot 3 + y_2 = 0$, $y_2 = \frac{3}{2}$; $-(\frac{2}{3}) \cdot (\frac{3}{2}) + y_3 = 2$, $y_3 = 3$. (3) Back substitution $Ux = y$: $(\frac{4}{3})x_3 = 3$, $x_3 = \frac{9}{4}$; $(\frac{3}{2})x_2 - \frac{9}{4} = \frac{3}{2}$, $x_2 = (\frac{15}{4}) \cdot (\frac{2}{3}) = \frac{5}{2}$; $2x_1 - \frac{5}{2} = 3$, $x_1 = \frac{11}{4}$. Thus, $x = [\frac{11}{4};\ 5/2;\ 9/4]$. Verification: $2 \cdot (\frac{11}{4}) - \frac{5}{2} = 3$; $-(\frac{11}{4}) + 2 \cdot (\frac{5}{2}) - \frac{9}{4} = 0$; $-(\frac{5}{2}) + 2 \cdot (\frac{9}{4}) = 2$. (4) Nearest neighbor structure: the matrix is tridiagonal (nonzero elements only on the main diagonal and immediately above and below it); x_2 interacts with both $x_1 and x_3$, there is no direct $x_1 - x_3$ connection ($a_{13} = a_{31} = 0$). Such a structure arises in heat conduction models along $arod$, spring chains, and one-dimensional finite difference equations (A is the standard discrete negative second derivative up to a grid step factor); elimination preserves the narrow structure (L and U are bidiagonal), so the system is solved cheaply. $(5)LDU$: $D = [200;\ 0\ 3/2\ 0;\ 00\frac{4}{3}] and D^{-1}U = [1 - \frac{1}{2}0;\ 0\ 1\ -2/3;\ 0\ 0\ 1]$, so $A = L \cdot D \cdot D^{-1}U$.

[Source: Week 6 Independent Study Assignment — 40 assigned problems (from basic 2×2 to challenge problems); instructions:
show elimination steps, verify products, and justify existence of inverse or decomposition. Solutions verified in Python]

Exercise 1. 2×2 **Inverse. Find** A^{-1} $for A=\begin{bmatrix}2&2\\1&6\end{bmatrix}$**. Verify by multiplication.**

Solution. $\det A = 2\cdot6-2\cdot1=10\neq0$. $A^{-1}=(\frac{1}{10})\cdot\begin{bmatrix}6&-2\\-1&2\end{bmatrix}=[\frac{3}{5}\ -\frac{1}{5};\ -1/10\ 1/5]$. Verification: $A\cdot A^{-1}=I$ — element $(1,1) is \frac{6}{5}-\frac{1}{5}=1$, element $(2,2) is -\frac{1}{5}+\frac{6}{5}=1$, off-diagonal elements are zeros.

Exercise 2. 2×2 **System Solution. Solve** $Ax=b$**, where** $A=\begin{bmatrix}3&1\\2&3\end{bmatrix}$, $b=\begin{bmatrix}8\\10\end{bmatrix}$**.**

Solution. System: 3x $+y=8$, 2x + 3y $=10$. Multiplying the first equation by 3 and subtracting the second, we get 7x $=14$, so $x=2, y=(8-6)/1=2$. Thus, $x=\begin{bmatrix}2\\2\end{bmatrix}$. Verification: $A\cdot x=\begin{bmatrix}6&+&2\\4&+&6\end{bmatrix}=\begin{bmatrix}8\\10\end{bmatrix}=b$.

Exercise 3. 2×2 **Inverse. Find** A^{-1} $for A=\begin{bmatrix}4&2\\1&4\end{bmatrix}$**. Verify by multiplication.**

Solution. $\det A = 4\cdot4-2\cdot1=14\neq0$. $A^{-1}=(\frac{1}{14})\cdot\begin{bmatrix}4&-2\\-1&4\end{bmatrix}=[\frac{2}{7}\ -\frac{1}{7};\ -1/14\ 2/7]$. Verification: $A\cdot A^{-1}=I$ — element $(1,1) is \frac{8}{7}-\frac{1}{7}=1$, element $(2,2) is -\frac{1}{7}+\frac{8}{7}=1$, off-diagonal elements are zeros.

Exercise 4. 2×2 **System Solution. Solve** $Ax=b$**, where** $A=\begin{bmatrix}2&1\\2&5\end{bmatrix}$, $b=\begin{bmatrix}4\\12\end{bmatrix}$**.**

Solution. System: 2x $+y=4$, 2x + 5y $=12$. Subtracting the first equation from the second: 4y $=8, y=2$; then 2x $=4-2, x=1$. Thus, $x=\begin{bmatrix}1\\2\end{bmatrix}$. Verification: $A\cdot x=\begin{bmatrix}2&+&2\\2&+&10\end{bmatrix}=\begin{bmatrix}4\\12\end{bmatrix}=b$.

Exercise 5. 2×2 **Decomposition. Find** $A=LU=LDU for A=\begin{bmatrix}2&1\\5&4\end{bmatrix}$**.**

Solution. $m_{21}=\frac{5}{2}$: $R_2\leftarrow R_2-(\frac{5}{2})R_1$ gives $U=[21;\ 0\ 3/2]$. $L=[10;\ \frac{5}{2}1], A=LU$ (verification: $(\frac{5}{2})\cdot\begin{bmatrix}2&1\end{bmatrix}+[0\frac{3}{2}]=\begin{bmatrix}5&4\end{bmatrix}$). $D=[20;\ 0\frac{3}{2}], D^{-1}U=[1\frac{1}{2};\ 0\ 1]$, and $A=L\cdot D\cdot D^{-1}U$.

Exercise 6. 2×2 **Decomposition. Find** $A=LU=LDU for A=\begin{bmatrix}3&1\\2&5\end{bmatrix}$**.**

Solution. $m_{21}=\frac{2}{3}$: $R_2\leftarrow R_2-(\frac{2}{3})R_1$ gives $U=[31;\ 0\frac{13}{3}](5-\frac{2}{3}=\frac{13}{3})$. $L=[10;\ \frac{2}{3}1], A=LU$ (verification: $(\frac{2}{3})\cdot\begin{bmatrix}3&1\end{bmatrix}+[0\frac{13}{3}]=\begin{bmatrix}2&5\end{bmatrix}$). $D=[30;\ 0\frac{13}{3}]$, $D^{-1}U=[1\frac{1}{3};\ 0\ 1]$, and $A=L\cdot D\cdot D^{-1}U$.

Exercise 7. 2×2 **Decomposition. Find** $A = LU = LDU for A = \begin{bmatrix} 1 & 1 \\ 3 & 5 \end{bmatrix}$**.**

Solution. $m_{21} = 3$: $R_2 \leftarrow R_2 - 3R_1$ gives $U = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$. $L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$, $A = LU$ (verification: $3 \cdot \begin{bmatrix} 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 5 \end{bmatrix}$). $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, $D^{-1}U = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, and $A = L \cdot D \cdot D^{-1}U$.

Exercise 8. 2×2 **Decomposition. Find** $A = LU = LDU for A = \begin{bmatrix} 2 & 1 \\ 4 & 4 \end{bmatrix}$**.**

Solution. $m_{21} = 2$: $R_2 \leftarrow R_2 - 2R_1$ gives $U = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$. $L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $A = LU$ (verification: $2 \cdot \begin{bmatrix} 2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 4 \end{bmatrix}$). $D = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, $D^{-1}U = [1\frac{1}{2}$; 0 1], and $A = L \cdot D \cdot D^{-1}U$.

Exercise 9. 3×3 **Elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix}$**.**

Solution. The augmented matrix is $\begin{bmatrix} 1 & 3 & 1 & 6 \\ 2 & 5 & 2 & 10 \\ 1 & 4 & 4 & 5 \end{bmatrix}$. Step $1 (m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, -1, 0| - 2)$; $R_3 - R_1 = (0, 1, 3| - 1)$. Step $2 (m_{32} = -1)$: $R_3 + R_2 = (0, 0, 3| - 3)$. The triangular system: $x +$ 3y $+z = 6, -y = -2,$ 3z $= -3$; hence $z = -1, y = 2, x = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 1 & + & 6 & 1 \\ 2 & + & 10 & 2 \\ 1 & + & 8 & 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix} = b.$

Exercise 10. 3×3 **Elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix}$**.**

Solution. The augmented matrix is $\begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 3 & 2 & 6 \\ 1 & 2 & 3 & 2 \end{bmatrix}$. Step $1 (m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, 1, 0|2)$; $R_3 - R_1 = (0, 1, 2|0)$. Step $2 (m_{32} = 1)$: $R_3 - R_2 = (0, 0, 2| - 2)$. The triangular system: $x + y + z = 2, y = 2,$ 2z $= -2$; hence $z = -1, y = 2, x = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$.

Verification: $A \cdot x = \begin{bmatrix} 1 & + & 2 & 1 \\ 2 & + & 6 & 2 \\ 1 & + & 4 & 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} = b.$

Exercise 11. 3×3 **Elimination (infinitely many solutions). Solve** $Ax = b$ **by**

Gaussian elimination for $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 3 & 6 \end{bmatrix}$, $b = \begin{bmatrix} 4 \\ 8 \\ 1 \end{bmatrix}$**.**

Solution. The augmented matrix is $\begin{bmatrix} 1 & 2 & 1 & 4 \\ 2 & 4 & 2 & 8 \\ 1 & 3 & 6 & 1 \end{bmatrix}$. Step $1(m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, 0, 0|0)$ — the second equation is twice the first and adds no constraints; $R_3 - R_1 = (0, 1, 5| - 3)$. After swapping rows $2and3$: $\begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & 1 & 5 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ — two pivots for three unknowns. Free variable $z = t$: $y = -3-$ 5t, $x = 4-$ 2y $-z = 10+$ 9t. Infinitely many solutions: $x = \begin{bmatrix} 10 & + & 9t \\ -3 & & 5t \\ t & & \end{bmatrix}$, $t \in R$; particular solution $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ corresponds to $t = -1$.

Verification: $A \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 & + & 4 & 1 \\ 2 & + & 8 & 2 \\ 1 & + & 6 & 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \\ 1 \end{bmatrix} = b.$

Exercise 12. 3×3 **Elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 10 \\ 6 \end{bmatrix}$**.**

Solution. The augmented matrix is $\begin{bmatrix} 1 & 3 & 1 & 6 \\ 2 & 5 & 2 & 10 \\ 1 & 4 & 3 & 6 \end{bmatrix}$. Step $1(m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, -1, 0| - 2)$; $R_3 - R_1 = (0, 1, 2|0)$. Step $2(m_{32} = -1)$: $R_3 + R_2 = (0, 0, 2| - 2)$. Back substitution: $z = -1, y = 2, x = 6 - 6 + 1 = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 1 & + & 6 & 1 \\ 2 & + & 10 & 2 \\ 1 & + & 8 & 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 6 \end{bmatrix} = b.$

Exercise 13. $LUandLDU3 \times 3$**. Compute** $LUandLDUforA = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 4 & 3 \\ -2 & 1 & 1 \end{bmatrix}$**.**

Solution. $m_{21} = 2, m_{31} = -1$: $R_2 - 2R_1 = (0, 2, 1)$; $R_3 + R_1 = (0, 2, 2)$. Second pivot $2, m_{32} = 1$: $R_3 - R_2 = (0, 0, 1)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$; verification: $LU = A$ (third

row: $-\begin{bmatrix}2 & 1 & 1\end{bmatrix}+\begin{bmatrix}0 & 2 & 1\end{bmatrix}+\begin{bmatrix}0 & 0 & 1\end{bmatrix}=\begin{bmatrix}-2 & 1 & 1\end{bmatrix}$). $D=\begin{bmatrix}2&0&0\\0&2&0\\0&0&1\end{bmatrix}$, $D^{-1}U=[1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/2; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Exercise 14. $LUandLDU3\times 3$**. Compute** $LUandLDUforA=\begin{bmatrix}2&1&1\\2&4&2\\-2&5&3\end{bmatrix}$**.**

Solution. $m_{21}=1, m_{31}=-1$: $R_2-R_1=(0,3,1)$; $R_3+R_1=(0,6,4)$. Second pivot $3, m_{32}=2$: $R_3-2R_2=(0,0,2)$. $L=\begin{bmatrix}1&0&0\\1&1&0\\-1&2&1\end{bmatrix}$, $U=\begin{bmatrix}2&1&1\\0&3&1\\0&0&2\end{bmatrix}$; verification: third $rowofLUis$ $-\begin{bmatrix}2 & 1 & 1\end{bmatrix}+2\cdot\begin{bmatrix}0 & 3 & 1\end{bmatrix}+\begin{bmatrix}0 & 0 & 2\end{bmatrix}=\begin{bmatrix}-2 & 5 & 3\end{bmatrix}$. $D=\begin{bmatrix}2&0&0\\0&3&0\\0&0&2\end{bmatrix}$, $D^{-1}U=[1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/3; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Exercise 15. $LUandLDU3\times 3$**. Compute** $LUandLDUforA=\begin{bmatrix}2&1&1\\4&6&3\\-2&11&5\end{bmatrix}$**.**

Solution. $m_{21}=2, m_{31}=-1$: $R_2-2R_1=(0,4,1)$; $R_3+R_1=(0,12,6)$. Second pivot $4, m_{32}=3$: $R_3-3R_2=(0,0,3)$. $L=\begin{bmatrix}1&0&0\\2&1&0\\-1&3&1\end{bmatrix}$, $U=\begin{bmatrix}2&1&1\\0&4&1\\0&0&3\end{bmatrix}$; verification: $-\begin{bmatrix}2 & 1 & 1\end{bmatrix}+3\cdot\begin{bmatrix}0 & 4 & 1\end{bmatrix}+\begin{bmatrix}0 & 0 & 3\end{bmatrix}=\begin{bmatrix}-2 & 11 & 5\end{bmatrix}$. $D=\begin{bmatrix}2&0&0\\0&4&0\\0&0&3\end{bmatrix}$, $D^{-1}U=[1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/4; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Exercise 16. $LUandLDU3\times 3$**. Compute** $LUandLDUforA=\begin{bmatrix}2&1&1\\2&3&2\\-2&1&1\end{bmatrix}$**.**

Solution. $m_{21}=1, m_{31}=-1$: $R_2-R_1=(0,2,1)$; $R_3+R_1=(0,2,2)$. Second pivot $2, m_{32}=1$: $R_3-R_2=(0,0,1)$. $L=\begin{bmatrix}1&0&0\\1&1&0\\-1&1&1\end{bmatrix}$, $U=\begin{bmatrix}2&1&1\\0&2&1\\0&0&1\end{bmatrix}$; verification: $LU=A$. $D=\begin{bmatrix}2&0&0\\0&2&0\\0&0&1\end{bmatrix}$, $D^{-1}U=[1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/2; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Exercise 17. Row Permutation and PLU**. Find** $arow$ **permutation** P **and factors satisfying** $PA=LU$**, for** $A=\begin{bmatrix}0&2&1\\1&1&0\\2&3&2\end{bmatrix}$**.**

Solution. Zero first pivot — swap rows $1 and 2$: $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $PA = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 2 & 3 & 2 \end{bmatrix}$. Elimination: $m_{31} = 2(R_3 - 2R_1 = (0,\ 1,\ 2))$; $m_{32} = \frac{1}{2}(R_3 - (\frac{1}{2})R_2 = (0,\ 0,\ \frac{3}{2}))$. $L = [100;\ 0\ 1\ 0;\ 2\frac{1}{2}1], U = [110;\ 0\ 2\ 1;\ 0\ 0\ 3/2]$; verification: $LU = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 2 & 3 & 2 \end{bmatrix} = PA$. Thus, $PA = LU$.

Exercise 18. Transpose and Inverse. For $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}$ compute A^T and verify $(A^T)^{-1} = (A^{-1})^T$.

Solution. $A^T = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$; $\det A = 1\cdot(1-0) - 2\cdot(0-2) + 0 = 5$. $(A^T)^{-1} = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix}$.

Independently: $A^{-1} = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{bmatrix}$, from which $(A^{-1})^T = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix} = (A^T)^{-1}$. The equality holds; verification: $A^T \cdot (A^T)^{-1} = I$.

Exercise 19. Symmetric LDL^T. Factor the symmetric $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix} as LDL^T$.

Solution. The multipliers are $m_{21} = 1$, $m_{31} = 0$, $m32 = \frac{1}{3}$. $L = [100;\ 1\ 1\ 0;\ 0\frac{1}{3}1], D = [200;\ 0\ 3\ 0;\ 0\ 0\ 8/3]$; for a symmetric matrix without pivoting, $A = LDL^T$. Verification: $L \cdot D \cdot L^T = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix} = A$.

Exercise 20. Challenge problem: tridiagonal model. Solve $Ax = b$ using $LU for A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$. Interpret the nearest neighbor structure.

Solution. LU: $m_{21} = -\frac{1}{2} and m_{32} = -\frac{2}{3}$, so $L = [100;\ -1/2\ 1\ 0;\ 0 - \frac{2}{3}1], U = [2 - 10;\ 0\ 3/2\ -1;\ 0\ 0\ 4/3]$. Forward substitution $Ly = b$: $y_1 = 2$, $y_2 = 1$, $y_3 = \frac{8}{3}$. Back substitution $Ux = y$: $x_3 = 2, x_2 = 2, x_1 = 2$. Thus, $x = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 4 & & 2 \\ -2 & + & 4 \\ -2 & + & 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = b$. The matrix is tridiagonal: connections only between nearest neighbors (x_2} with $x_1 and x_3$; no direct connection $x_1 - x_3$, $a_{13} = a_{31} = 0$); $L and U$ are bidiagonal — the narrow structure is preserved.

Exercise 21. 2×2 inverse. Find $A^{-1} for A = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$. Verify by multiplication.

Solution. $\det A = 2\cdot 3 - 1\cdot 2 = 4 \neq 0$. $A^{-1} = (\frac{1}{4})\cdot \begin{bmatrix} 3 & -1 \\ -2 & 2 \end{bmatrix} = [\frac{3}{4} -\frac{1}{4};\ -1/2\ 1/2]$. Verification: $A \cdot A^{-1} = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix} \cdot [\frac{3}{4} - \frac{1}{4};\ -\frac{1}{2}\frac{1}{2}] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$.

Exercise 22. Solving $a2 \times 2$ **system. Solve** $Ax = b$**, where** $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 8 \\ 6 \end{bmatrix}$**.**

Solution. System: 3x + 2y $= 8, x+$ 4y $= 6$. From the second equation $x = 6-$ 4y; substitution: $3(6 - 4y)+$ 2y $= 8$, that is 18 − 10y $= 8, y = 1, x = 2$. Thus, $x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 6 & + & 2 \\ 2 & + & 4 \end{bmatrix} = \begin{bmatrix} 8 \\ 6 \end{bmatrix} = b$.

Exercise 23. 2×2 **inverse. Find** $A^{-1} for A = \begin{bmatrix} 4 & 1 \\ 2 & 5 \end{bmatrix}$**. Verify by multiplication.**

Solution. $\det A = 4 \cdot 5 - 1 \cdot 2 = 18 \neq 0$. $A^{-1} = (\frac{1}{18}) \cdot \begin{bmatrix} 5 & -1 \\ -2 & 4 \end{bmatrix} = [\frac{5}{18} - \frac{1}{18};\ -1/9\ 2/9]$. Verification: $A \cdot A^{-1} = I$ — element (1,1) equals $\frac{20}{18} - \frac{2}{18} = 1$, element (2,2) equals $-\frac{2}{18} + \frac{20}{18} = 1$, off-diagonal elements are zeros.

Exercise 24. Solving $a2 \times 2$ **system. Solve** $Ax = b$**, where** $A = \begin{bmatrix} 2 & 2 \\ 1 & 6 \end{bmatrix}$, $b = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$**.**

Solution. System: 2x + 2y $= 4$ (that is, $x + y = 2), x+$ 6y $= 7$. Subtracting: 5y $= 5, y = 1, x = 1$. Thus, $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 2 & + & 2 \\ 1 & + & 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix} = b$.

Exercise 25. 2×2 **Factorization. Find** $A = LU = LDU for A = \begin{bmatrix} 2 & 1 \\ 2 & 4 \end{bmatrix}$**.**

Solution. $m_{21} = 1$: $R_2 \leftarrow R_2 - R_1$ gives $U = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$. $L = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $A = LU$ (verification: $\begin{bmatrix} 2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 4 \end{bmatrix}$). $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, $D^{-1}U = [1\frac{1}{2};\ 0\ 1]$, and $A = L \cdot D \cdot D^{-1}U$.

Exercise 26. 2×2 **Factorization. Find** $A = LU = LDU for A = \begin{bmatrix} 3 & 1 \\ 3 & 5 \end{bmatrix}$**.**

Solution. $m_{21} = 1$: $R_2 \leftarrow R_2 - R_1$ gives $U = \begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix}$. $L = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $A = LU$ (verification: $\begin{bmatrix} 3 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 5 \end{bmatrix}$). $D = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$, $D^{-1}U = [1\frac{1}{3};\ 0\ 1]$, and $A = L \cdot D \cdot D^{-1}U$.

Exercise 27. 2×2 **Factorization. Find** $A = LU = LDU for A = \begin{bmatrix} 1 & 1 \\ 4 & 3 \end{bmatrix}$**.**

Solution. $m_{21} = 4$: $R_2 \leftarrow R_2 - 4R_1$ gives $U = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$. $L = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$, $A = LU$ (verification: $4 \cdot \begin{bmatrix} 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -1 \end{bmatrix} = \begin{bmatrix} 4 & 3 \end{bmatrix}$). $D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ (negative pivot is acceptable), $D^{-1}U = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, and $A = L \cdot D \cdot D^{-1}U$.

Exercise 28. 2×2 **Factorization. Find** $A = LU = LDU for A = \begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix}$.

Solution. $m_{21} = \frac{5}{2}$: $R_2 \leftarrow R_2 - (\frac{5}{2})R_1$ gives $U = [21;\ 0\ 3/2]$. $L = [10;\ \frac{5}{2}1], A = LU$ (verification: $(\frac{5}{2}) \cdot \begin{bmatrix} 2 & 1 \end{bmatrix} + [0\frac{3}{2}] = \begin{bmatrix} 5 & 4 \end{bmatrix}$). $D = [20; 0\frac{3}{2}], D^{-1}U = [1\frac{1}{2};\ 0\ 1]$, and $A = L \cdot D \cdot D^{-1}U$.

Exercise 29. 3×3 **Elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 10 \\ 6 \end{bmatrix}$.

Solution. Augmented matrix $\left[\begin{array}{ccc|c} 1 & 3 & 1 & 6 \\ 2 & 5 & 2 & 10 \\ 1 & 4 & 3 & 6 \end{array}\right]$. Step $1 (m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, -1, 0 | -2)$; $R_3 - R_1 = (0, 1, 2 | 0)$. Step $2 (m_{32} = -1)$: $R_3 + R_2 = (0, 0, 2 | -2)$. Back substitution: $z = -1, y = 2, x = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 6 \\ 10 \\ 6 \end{bmatrix} = b$.

Exercise 30. 3×3 **Elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix}$.

Solution. The augmented matrix is $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 3 & 2 & 6 \\ 1 & 2 & 4 & 1 \end{array}\right]$. Step $1 (m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, 1, 0 | 2)$; $R_3 - R_1 = (0, 1, 3 | -1)$. Step $2 (m_{32} = 1)$: $R_3 - R_2 = (0, 0, 3 | -3)$. Back substitution: $z = -1, y = 2, x = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix} = b$.

Exercise 31. 3×3 **Elimination (infinitely many solutions). Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 3 & 5 \end{bmatrix}$, $b = \begin{bmatrix} 4 \\ 8 \\ 2 \end{bmatrix}$.

Solution. The augmented matrix is $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 4 & 2 & 8 \\ 1 & 3 & 5 & 2 \end{array}\right]$. Step $1 (m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, 0, 0 | 0)$ — the second equation is twice the first; $R_3 - R_1 = (0, 1, 4 | -2)$. After swapping

rows $2 and 3$: $\begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & 1 & 4 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. Free variable $z = t$: $y = -2-$ 4t, $x = 4-$ 2y $-z = 8+$ 7t. Infinitely many solutions: $x = \begin{bmatrix} 8 & + & 7t \\ -2 & & 4t \\ t & & \end{bmatrix}$, $t \in R$; particular solution $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ ($t = -1$). Verification: $A \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \\ 2 \end{bmatrix} = b$.

Exercise 32. 3×3 **Elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix}$.

Solution. The augmented matrix is $\begin{bmatrix} 1 & 3 & 1 & 6 \\ 2 & 5 & 2 & 10 \\ 1 & 4 & 4 & 5 \end{bmatrix}$. Step $1 (m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, -1, 0 | -2)$; $R_3 - R_1 = (0, 1, 3 | -1)$. Step $2 (m_{32} = -1)$: $R_3 + R_2 = (0, 0, 3 | -3)$. Back substitution: $z = -1, y = 2, x = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix} = b$.

Exercise 33. $LU and LDU 3 \times 3$**. Compute** $LU and LDU for A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ -2 & 1 & 1 \end{bmatrix}$.

Solution. $m_{21} = 1, m_{31} = -1, m_{32} = 1$: $R_2 - R_1 = (0, 2, 1)$; $R_3 + R_1 = (0, 2, 2)$; $R_3 - R_2 = (0, 0, 1)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$; verification: $LU = A$. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $D^{-1}U = [1 \frac{1}{2} \frac{1}{2}$; 0 1 1/2; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Exercise 34. $LU and LDU 3 \times 3$**. Compute** $LU and LDU for A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 5 & 3 \\ -2 & 5 & 3 \end{bmatrix}$.

Solution. $m_{21} = 2$, $m_{31} = -1$, $m_{32} = 2$: $R_2 - 2R_1 = (0, 3, 1)$; $R_3 + R_1 = (0, 6, 4)$; $R_3 - 2R_2 = (0, 0, 2)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$; verification: $LU = A$ (third row: $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + 2 \cdot \begin{bmatrix} 0 & 3 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 5 & 3 \end{bmatrix}$). $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, $D^{-1}U = [1 \frac{1}{2} \frac{1}{2}$; 0 1 1/3; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Exercise 35. $LU and LDU 3\times 3$**. Compute** $LU and LDU for A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 5 & 2 \\ -2 & 11 & 5 \end{bmatrix}$**.**

Solution. $m_{21}=1$, $m_{31}=-1$, $m_{32}=3$: $R_2-R_1=(0,4,1)$; $R_3+R_1=(0,12,6)$; $R_3-3R_2=(0,0,3)$. $L=\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix}$, $U=\begin{bmatrix} 2 & 1 & 1 \\ 0 & 4 & 1 \\ 0 & 0 & 3 \end{bmatrix}$; verification: $LU=A$ (third row: $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix}+3\cdot\begin{bmatrix} 0 & 4 & 1 \end{bmatrix}+\begin{bmatrix} 0 & 0 & 3 \end{bmatrix}=\begin{bmatrix} -2 & 11 & 5 \end{bmatrix}$). $D=\begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, $D^{-1}U=[1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/4; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Exercise 36. $LU and LDU 3\times 3$**. Compute** $LU and LDU for A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 4 & 3 \\ -2 & 1 & 1 \end{bmatrix}$**.**

Solution. $m_{21}=2$, $m_{31}=-1$, $m_{32}=1$: $R_2-2R_1=(0,2,1)$; $R_3+R_1=(0,2,2)$; $R_3-R_2=(0,0,1)$. $L=\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$, $U=\begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$; verification: $LU=A$. $D=\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $D^{-1}U=[1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/2; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Exercise 37. Row Permutation and PLU**. Find** $a row$ **permutation** P **and factors satisfying** $PA=LU$**, for** $A=\begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 3 & 2 \end{bmatrix}$**.**

Solution. Swap rows $1 and 2$ (zero first pivot): $P=\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $PA=\begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 2 & 3 & 2 \end{bmatrix}$. Elimination: $m_{31}=2(R_3-2R_1=(0,\,1,\,2))$; $m_{32}=\frac{1}{2}(R_3-(\frac{1}{2})R_2=(0,\,0,\,\frac{3}{2}))$. $L=[100$; 0 1 0; $2\frac{1}{2}1], U=[110$; 0 2 1; 0 0 3/2]; verification: $LU=PA$. Thus, $PA=LU$.

Exercise 38. Transpose and Inverse. For $A=\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}$ **compute** A^T **and verify** $(A^T)^{-1}=(A^{-1})^T$**.**

Solution. $A^T=\begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$; $\det A=5$. $(A^T)^{-1}=(\frac{1}{5})\cdot\begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix}$; on the other hand, $A^{-1}=(\frac{1}{5})\cdot\begin{bmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{bmatrix}$, and $(A^{-1})^T=(\frac{1}{5})\cdot\begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix}=(A^T)^{-1}$. The equality holds; verification: $A^T\cdot(A^T)^{-1}=I$.

Exercise 39. Symmetric LDL^T. Factor the symmetric $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix}$ as LDL^T.

Solution. Multipliers $m_{21} = 1$, $m_{31} = 0$, $m_{32} = \frac{1}{3}$. $L = [100; 1\ 1\ 0; 0\ \frac{1}{3}\ 1], D = [200; 0\ 3\ 0; 0\ 0\ 8/3]$, and $A = LDL^T$. Verification: $L \cdot D \cdot L^T = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix} = A$.

Exercise 40. Challenge problem: tridiagonal model. Solve $Ax = b$ using LU for $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$. Interpret the nearest neighbor structure.

Solution. LU: $L = [100; -1/2\ 1\ 0; 0\ -\frac{2}{3}\ 1], U = [2\ -1\ 0; 0\ 3/2\ -1; 0\ 0\ \frac{4}{3}]$ (*the* multipliers $m_{21} = -\frac{1}{2}$, $m_{32} = -\frac{2}{3}$ are stored in L with their original signs). Forward substitution $Ly = b$ gives $y = [2; 1; 8/3]$; backward substitution $Ux = y$ gives $x_3 = 2, x_2 = 2, x_1 = 2$. Thus, $x = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = b$. The matrix is tridiagonal — each unknown is connected only to its nearest neighbors; the factors L and U are bidiagonal, and the narrow structure makes the solution inexpensive.

[Source: Week 6 Extra material — Review with examples: *I*) a brief lesson on LU, LDU, and normalized U; *II*) a brief lesson on

Gaussian elimination for 3×3 systems; *III*) a detailed example of finding the inverse by the Gauss-Jordan method]

Key points of lesson *II* (review of week 5). A system can be viewed in three pictures: *row* (equations are planes), matrix ($Ax = b$), and column (combination of columns of A gives b). Three reversible *row* operations are allowed: swapping two equations ($R_i \leftrightarrow R_j$), multiplying an equation by a nonzero number ($R_i \leftarrow c \cdot R_i, c \neq 0$), and adding a multiple of another equation ($R_i \leftarrow R_i + c \cdot R_j$); each is applied to the entire equation, including the right-hand side. Procedure for $a 3 \times 3$ system: write the augmented matrix; select a nonzero first pivot element (swapping rows if necessary); eliminate elements below it; select the second pivot element in the updated matrix; eliminate the element below it; check for contradictions or free variables; with three pivot elements, perform backward substitution; verify the solution in the original system. Backward substitution formulas: $z = c_3/u_{33}$, $y = (c_2 - u_{23} \cdot z)/u_{22}$, $x = (c_1 - u_{12} \cdot y - u_{13} \cdot z)/u_{11}$ (pivot elements are nonzero). Typical errors: forgetting to update the right-hand side; using the old *row* after it has been changed; confusing the direction of the multiplier relationship; considering a zero proposed pivot element as proof of degeneracy.

Example 1. LU, normalization, and LDU (lesson I). Find the LU and LDU decompositions of the matrix $A = \begin{bmatrix} 2 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{bmatrix}$.

Solution. Eliminate under the first pivot element 2: $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$

give $\begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 3 \\ 0 & 3 & 6 \end{bmatrix}$; then under the second pivot element 3: $R_3 \leftarrow R_3 - R_2$ gives $\begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{bmatrix}$. The multipliers are $m_{21} = 2$, $m_{31} = 1$, $m_{32} = 1$, so $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} andA = LU$ with $U = \begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{bmatrix}$; L stores the multipliers with their original signs (the operation subtracts $2R_1$, but $L_{21} = +2$). Normalization: divide each $rowofUby$ its diagonal element — the pivot elements 2, 3, 3 are factored out into $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, the normalized upper factor is $D^{-1}U = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, and $A = L \cdot D \cdot (D^{-1}U)$; multiplication by $D^{-1}on$ the left scales the rows. Note on conventions: in $A = LU$, the pivot elements sit on the diagonal of $U(Disnot$ separated, L has unit diagonal); in $A = LDU$, the pivot elements are in D, and both $Land$ the third factor have unit diagonals; many textbooks rename $D^{-1}UasU$.

Example 2. Solving a system using factors (lesson I). Using the decomposition from example 1, solve $Ax = \begin{bmatrix} 8 \\ 22 \\ 17 \end{bmatrix}$.

Solution. Forward substitution $Ly = b$: $y_1 = 8$; $2y_1 + y_2 = 22$, from which $y_2 = 6$; $y_1 + y_2 + y_3 = 17$, from which $y_3 = 3$, that is, $y = \begin{bmatrix} 8 \\ 6 \\ 3 \end{bmatrix}$. Backward substitution $Ux = y$: $3x_3 = 3, x_3 = 1$; $3x_2 + 3x_3 = 6, x_2 = 1$; $2x_1 + 4x_2 + 2x_3 = 8$, $x_1 = 1$. Thus, $x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. With LDU, the computation breaks into three steps: $Ly = b, Dz = y$ (division by pivot elements), then $(D^{-1}U)x = z$. When row exchanges are needed, the standard form is $PA = LUorPA = LDU$, where P records the exchanges; extraction and division by pivot elements requires them to be nonzero.

Example 3. Complete Gaussian elimination example (lesson II). Solve the system 2x $+y - z = 1$, 4x + 3y $+z = 13, -$2x $+y+$ 2z $= 6$.

Solution. Here $A = \begin{bmatrix} 2 & 1 & -1 \\ 4 & 3 & 1 \\ -2 & 1 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 13 \\ 6 \end{bmatrix}$. Step 1: the first pivot element is 2, multiplier $m_{21} = \frac{4}{2} = 2$; $R_2 \leftarrow R_2 - 2R_1$: $(4x + 3y + z) - 2(2x + y - z) = 13 - 2 \cdot 1$, that is, $y+$ 3z $= 11$. Step 2: $m_{31} = -\frac{2}{2} = -1$; $R_3 \leftarrow R_3 + R_1$: $(-2x + y + 2z) + (2x + y - z) = 6 + 1$, that is, 2y $+z = 7$. Step 3: the second pivot element is 1 (coefficient of yin the updated equation $2), m_{32} = 2$; $R_3 \leftarrow R_3 - 2R_2$: $(2y + z) - 2(y + 3z) = 7 - 22$, that is, $-$5z $= -15$.

The pivot elements 2, 1, -5 — the second and third are taken from the updated equations, not from the original diagonal. Back substitution: $z = 3, y = 2, x = 1$, that is, $(x, y, z) = (1, 2, 3)$. Verification: $2 \cdot 1 + 2 - 3 = 1$; $4 \cdot 1 + 6 + 3 = 13$; $-2 \cdot 1 + 2 + 6 = 6$. With the augmented matrix: $\begin{bmatrix} 2 & 1 & -1 & 1 \\ 4 & 3 & 1 & 13 \\ -2 & 1 & 2 & 6 \end{bmatrix} \to \begin{bmatrix} 2 & 1 & -1 & 1 \\ 0 & 1 & 3 & 11 \\ 0 & 2 & 1 & 7 \end{bmatrix} \to \begin{bmatrix} 2 & 1 & -1 & 1 \\ 0 & 1 & 3 & 11 \\ 0 & 0 & -5 & -15 \end{bmatrix}$.

Example 4. Elimination as matrix multiplication (Lesson II). Write the operations of Example 3 in terms of elimination matrices.

Solution. For $R_2 \leftarrow R_2 - 2R_1$: $E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (the second $rowof$ the product $E_{21} \cdot A$ equals $-2 \cdot (row1ofA) + (row2ofA)$); for adding $row1torow3$: $E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$; for $R_3 \leftarrow R_3 - 2R_2$: $E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$. Complete elimination: $E_{32} \cdot E_{31} \cdot E_{21} \cdot A = UandE_{32} \cdot E_{31} \cdot E_{21} \cdot b = c$, where $U = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & -5 \end{bmatrix}$, $c = \begin{bmatrix} 1 \\ 11 \\ -15 \end{bmatrix}$; the rightmost operation acts first. Rules: $(BC)_{ij} = \sum_k b_{ik} \cdot c_{kj}$; associativity $(BC)D = B(CD)$; non-commutativity $BC \neq CB$ — therefore the order of row operations is important.

Example 5. Zero proposed pivot element (Lesson II). Solve the system $y + z = 5$, $x + y + z = 6$, $\mathbf{2x + 3y + 4z = 20}$.

Solution. $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 2 & 3 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 5 \\ 6 \\ 20 \end{bmatrix}$; the first proposed pivot element is zero — we swap rows $1and2$ (permutation matrix $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, reordered system PAx $= Pb$): $\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 5 \\ 2 & 3 & 4 & 20 \end{bmatrix}$. Then $R_3 \leftarrow R_3 - 2R_1$: (0, 1, 2 | 8); then $R_3 \leftarrow R_3 - R_2$: (0, 0, 1 | 3). Back substitution: $z = 3, y = 2, x = 1$. A zero in the proposed pivot position does not automatically mean no unique solution; for numerical calculations, partial pivoting takes the largest available element in absolute value in the current column, controlling the elimination multipliers.

Example 6. How many solutions does the system have (Lesson II). Consider the family $x + y + z = 6$, $\mathbf{2x + 3y + 4z} = 20$, $\mathbf{3x + 4y + 5z} = k$. For which k are there no solutions, for which are there infinitely many?

Solution. Apply $R_2 \leftarrow R_2 - 2R_1 and R_3 \leftarrow R_3 - 3R_1$: we get $y + 2z = 8 and y + 2z = k - 18$. Subtracting equation 2 from equation 3: $0 = k - 26$. $If k \neq 26$ — no solutions (when

$k = 27weget0 = 1$ — the equations are inconsistent). $Ifk = 26$ — the last equation is $0 = 0$ and adds no constraints: with free variable $z = t$ we have $y = 8-$ 2t and $x = 6 - y - z = t - 2$, that is, $(x, y, z) = (t - 2, 8 - 2t, t), t \in R$ — infinitely many solutions. Outcome table for 3×3: three pivot elements — unique solution; $arow[000|d], d \neq 0$ — no solutions; fewer than three pivot elements without a contradictory row — infinitely many solutions (one zero row by itself does not guarantee consistency — need to check the other rows).

Example 7. LU**-factorization and reuse (Lesson** II**). Write** $A = LUfor$ **the system of Example 3 and solve** $Lc = b, Ux = c$**.**

Solution. The multipliers of Example 3: $m_{21} = 2$, $m_{31} = -1$, $m_{32} = 2$; we place them under the diagonal: $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$, and $A = LU$ with $U = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & -5 \end{bmatrix}$; the signs in $Lare$ the multipliers themselves, while the elimination operations use their negatives. Solve in two steps: $Lc = b$ gives $c_1 = 1$, $c_2 = 11$, $c_3 = -15$ (forward substitution); then $Ux = c$ by back substitution gives (1, 2, 3). The advantage appears when several systems have the same matrix A but different right-hand sides: A is factored once, and $LandU$ are reused. $Ifrow$ exchanges are needed, the standard factorization is $PA = LU$.

Example 8. Exercises $AandB$ **(Lesson** II**). Solve the systems:** $(A)x + y + z = 6$**, 2x + 3y +**$z = 11, x+$ **2y + 3z =** 14**;** $(B)y+$ **2z =** $8, x + y + z = 6$**, 2x + 3y +**$z = 11$**.**

Solution. (A) Eliminate x: R_2-2R_1 gives $y-z = -1$; R_3-R_1 gives $y+$ 2z $= 8$. Subtracting: 3z $= 9$, $z = 3$, $y = z - 1 = 2$, $x = 6 - 2 - 3 = 1$. Answer (1, 2, 3); check: $1 + 2 + 3 = 6$, $2+6+3 = 11$, $1+4+9 = 14$. (B) There is no xin the first equation (zero proposed pivot element) — before eliminating x, swap the first two rows: $x+y+z = 6, y+$ 2z $= 8$, 2x + 3y $+z = 11$. Eliminate x: R_3-2R_1 gives $y-z = -1$; together with $y+$ 2z $= 8$, subtraction gives 3z $= 9, z = 3, y = 2, x = 1$. Answer (1, 2, 3); check: $2 + 2 \cdot 3 = 8$, $1 + 2 + 3 = 6$, $2 + 6 + 3 = 11$.

Example 9. Exercises $CandD$ **(Lesson** II**).** (C) **Determine for which** k **the system** $x + y + z = 6$**, 2x + 3y + 4z =** 20**, 3x + 4y + 5z =** $khas$ **no solutions, and for which it has infinitely many.** (D) **Find the matrix of the operation** $R_3 \leftarrow R_3 - 4R_1$**.**

Solution. (C) This is a family of example 6: after elimination, the third equation reduces to $0 = k - 26$, so when $k \neq 26$ there are no solutions, when $k = 26$ there are infinitely many: $(x, y, z) = (t-2, 8-2t, t), t \in R$; there is no value of k that gives a unique solution (the coefficient matrix has only two pivots). (D) The operation changes only $row3$, subtracting $4 \cdot (row1)$: $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$; the third $rowof$ the product $E \cdot Ais(row3ofA) - 4 \cdot (row1ofA)$, the other rows do not change.

Example 10. Inverse matrix by the Gauss-Jordan method (Lesson III**). Find the inverse matrix of** $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$**, by adjoining the identity matrix and reducing the left part to the identity.**

Solution. Theory: the inverse of an n×n matrix Ais the matrix A^{-1} with $AA^{-1} = A^{-1}A = I$; each elementary row operation is equivalent to left multiplication by an elementary matrix, and if $E_k \cdots E_2 E_1 A = I$, then $A^{-1} = E_k \cdots E_2 E_1$ — therefore the same operations, applied to I, give A^{-1}: $[A|I] \to [I|A^{-1}]$; each operation acts on the entire augmented row. Step 1: $[A|I] = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{bmatrix}$. Step 2 (eliminate under the first pivot): $R_2 \leftarrow R_2 - R_1$: $\begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{bmatrix}$. Step 3 (under the second): $R_3 \leftarrow R_3 - R_2$: $\begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{bmatrix}$. Step 4 (above the third): $R_2 \leftarrow R_2 - R_3$: $\begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{bmatrix}$. Step 5 (above the second): $R_1 \leftarrow R_1 - R_2$: $\begin{bmatrix} 1 & 0 & 0 & 3 & -2 & 1 \\ 0 & 1 & 0 & -2 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{bmatrix}$. The left part became the identity, so $A^{-1} = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{bmatrix}$. Check by multiplication: $A \cdot A^{-1} = I(for$ example, element (2, 1): $1 \cdot 3 + 2 \cdot (-2) + 1 \cdot 1 = 0)$.

Example 11. When the method doesn't work (Lesson III). Show that the matrix $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ has no inverse.

Solution. A square matrix is invertible only if the left block in $[A|I]$ reduces to the identity matrix. Apply $R_2 \leftarrow R_2 - 2R_1$: $\begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{bmatrix}$. $Arow$ of zeros appeared in the left block — there are too few pivots to form I; the matrix is degenerate (singular), no inverse exists. Note: Gaussian elimination stops at an upper triangular matrix, while Gauss-Jordan continues until each pivot is $1and$ all elements above and below each pivot are zero.

4 Week 7 — Transposes, Permutations and Vector Spaces (Linear Algebra I)

[Source: Week 7 lecture "Transposes, permutations, vector spaces and subspaces" (Strang, Introduction to Linear Algebra, 4th ed.,
§2.7 pp. 108–120, §3.1 pp. 121–132)]

4.5.1 Review of Week 6: $A = LU, LDU, PA = LU$

Gaussian elimination gives $E_k \cdots E_1 A = U$. Without row exchanges, the elimination factors combine into a lower triangular L (with ones on the diagonal), and $A = LU$: for example, $A = \begin{bmatrix} 2 & 1 \\ 4 & 5 \end{bmatrix}$, the step $R2 \leftarrow R2 - 2R1$ gives $U = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$, $L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$. Factoring

out the pivots gives $A = \text{LD}\bar{\text{U}}$: $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} is$ the diagonal of pivots, and $\bar{\text{U}}$ (U with $abar) = [1\frac{1}{2}$; 0 1] is upper triangular with ones on the diagonal.

$$A = LU, A = \text{LD}\bar{\text{U}}; whenexchangingrows—PA = LU$$

If the leading element is zero ($A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$), the rows are swapped: $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ gives $PA = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$; all row swaps are recorded in P.

4.5.2 Transposition: definition and basic identities

$IfAisanm \times n$ matrix, then A^T has size $n \times m$ and elements $(A^T)_{ij} = a_{ji}$: columns become rows. Example: $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ gives $A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$.

$$(A^T)^T = A, (A + B)^T = A^T + B^T, (cA)^T = cA^T$$

Element-wise proof ($forthesum$): $((A + B)^T)_{ij} = (A + B)_{ji} = a_{ji} + b_{ji} = (A^T + B^T)_{ij}$.

$$(AB)^T = B^T A^T$$

The order of factors is reversed (A^{}{T} B^{}{T} may not even be defined). Proof: $((AB)^T)_{ij} = (AB)_{ji} = \Sigma_k a_{jk} b_{ki} = \Sigma_k (B^T)_{ik} (A^T)_{kj} = (B^T A^T)_{ij}$. Lecture check: $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix}$: $(AB)^T = \begin{bmatrix} 8 & 3 \\ 2 & 1 \end{bmatrix} = B^T A^T$.

$$(A^{-1})^T = (A^T)^{-1}$$

For an invertible A, we transpose $AA^{-1} = I$: we get $(A^{-1})^T A^T = I$; similarly $A^T (A^{-1})^T = I$. Thus, transposition and inversion commute.

4.5.3 Symmetric and skew-symmetric matrices; the product $A^T A$

A matrix S is symmetric if $S^T = S (s_{ij} = s_{ji})$; example: $S = \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}$. A matrix K is skew-symmetric if $K^T = -K$; example: $K = \begin{bmatrix} 0 & 4 \\ -4 & 0 \end{bmatrix}$; the diagonal is zero: $k_{ii} = -k_{ii} \Rightarrow k_{ii} = 0$.

$$A = (A + A^T)/2 + (A - A^T)/2 = S + K, S^T = S, K^T = -K$$

Any square matrix decomposes this way; lecture example: $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \to S = \begin{bmatrix} 1 & 3 \\ 3 & 3 \end{bmatrix}$, $K = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$; verification: $S + K = A$.

$$(A^TA)^T = A^TA, x^TA^TAx = (Ax)^T(Ax) = ||Ax||^2 \geq 0$$

The product A^TA is always symmetric (a special case of $(AB)^T = B^TA^T$), and $x^TA^TAx = ||Ax||^2 \geq 0 for any x$. Example: $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ gives $A^TA = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$; matrices R^TR are fundamental to least squares. Scalar and outer product: for columns $u, v \in R^n$, the number $u^Tv is$ the scalar product, while $uv^T is an \times n$ matrix. Example: $u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $v = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$: $u^Tv = 11$, $uv^T = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$.

4.5.4 Permutations: $P^T = P^{-1}$, six 3×3 matrices, partial pivoting

A permutation matrix is obtained by permuting the rows of the identity matrix: exactly one 1 in each row and each column. Example: $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ maps $[x_1; x_2; x_3] to [x_2; x_3; x_1]$.

$$PA—matrix A with permuted rows; AQ—with permuted columns$$

Left multiplication permutes rows: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$, $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \to PA = \begin{bmatrix} 3 & 4 \\ 5 & 6 \\ 1 & 2 \end{bmatrix}$. Right multiplication permutes columns: $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, $Q = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \to AQ = \begin{bmatrix} 2 & 3 & 1 \\ 5 & 6 & 4 \end{bmatrix}$. In general, $PA \neq AP$.

$$P^TP = I, i.e. P^{-1} = P^T; consequences: ||Px|| = ||x||, \det P = \pm 1$$

The columns of P are different coordinate unit vectors, so their scalar products form I: $P^TP = I$, hence $P^{-1} = P^T$. Consequences: $\|Px\| = \|x\|$ and $\det P = \pm 1$. One row exchange changes the sign of the determinant: $\det \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = -1$, so $\det(PA) = -\det A$; two exchanges restore the sign. There are exactly $n!$ permutations of order n; for $n = 3$ there are six (Strang, §2.7) — three even ($\det P = +1$) and three odd ($\det P = -1$); $P^{-1} = P^T$, so $(P_{32}P_{21})^{-1} = P_{21}P_{32}$:

Permutation	**Matrix (row-wise)**	$\det P$	$P^{-1} = P^T$
I	$\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$	$+1$	I
P_{21} (swap rows 1, 2)	$\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}$	-1	P_{21}
P_{32} (swap rows 2, 3)	$\begin{bmatrix}1&0&0\\0&0&1\\0&1&0\end{bmatrix}$	-1	P_{32}
P_{31} (swap rows 1, 3)	$\begin{bmatrix}0&0&1\\0&1&0\\1&0&0\end{bmatrix}$	-1	P_{31}
$P_{32}P_{21}$ (cyclic)	$\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}$	$+1$	$P_{21}P_{32}$
$P_{21}P_{32}$ (cyclic)	$\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$	$+1$	$P_{32}P_{21}$

$P32P21$ (cyclic) $\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix} + 1P21P32P21P32$(cyclic) $\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix} + 1P32P21$

Partial pivoting: if the pivot element is zero, rows are swapped, taking in the current column the candidate with the largest absolute value. Example: $A = [10^{-5}1;\ 1\ 1]$: without swap the multiplier is $\frac{1}{10}^{-5} = 10^5$, after swap — $10^{-5}/1 = 10^{-5}$. Swaps are recorded in P: $PA = LU$.

4.5.5 Vector spaces: definition and examples

Vector space over R — $aset$ of vectors with addition and scalar multiplication operations, satisfying vector space axioms: zero vector, opposite elements, associativity, distributivity, etc. Lecture examples: R^n, real matrices, polynomials, continuous functions — any object can be a "vector".

4.5.6 Subspaces and closure testing

$$W \subseteq V — subspace \Leftrightarrow 0 \in W; u + v \in W for all u, v \in W; cu \in W for all u \in W, c \in R$$

Equivalently: non-empty W is closed under combinations au + bv. Test: (1) zero vector; (2) sum; (3) scalar multiplication — violation of any point refutes.

- Line through origin: $W = \{(t,\ 2t)\} = \text{span}\{(1,\ 2)\}$: for $u = (s,\ 2s), v = (t,\ 2t)$

the combination au $+bv = (as + bt, 2(as + bt))$ is again in W — subspace.

- Shifted line (counterexample): $S = \{(x, y)$: $y = 2x + 1\}$ — zero vector does

not satisfy the equation; a subspace must pass through the origin.

- Plane through origin: $W = \{(x, y, z)\colon\ x + y + z = 0\} = \text{span}\{(1,\ 0,\ -1),\ (0,\ 1,$

$-1)\}$ (from $z = -x - y$: $(x,\ y,\ z) = x(1,\ 0,\ -1) + y(0,\ 1,\ -1)$); plane $x + y + z = 1 isnota$ subspace (no zero).

- Geometry of R^3: subspaces are only $\{0\}$, lines through 0, planes through 0

and all R^3.

- Intersection and union: $U \cap W$ — subspace; union — not necessarily: $xandy$

axes are subspaces of R^2, but $(1, 0) + (0, 1) = (1, 1)$ lies on neither axis.

- Matrices and polynomials: symmetric 2×2 — subspace of all 2×2 (sums and

multiples are symmetric); polynomials with $p(0) = 0$ — subspace, with $p(0) = 1$ — no.

4.5.7 Linear combinations and linear spans

$$\text{span} v_1, ..., v_k = c_1 v_1 + ... + c_k v_k : c_i \in R$$

Linear span — the set of all combinations of given vectors; it is always a subspace: if $u = \Sigma a_i v_i and v = \Sigma b_i v_i$, then su $+tv = \Sigma(sa_i + tb_i)v_i$ — again a combination. Lecture example: is $(3, 5) in$span$\{(1,\ 1),\ (1,\ -1)\}$? Solve $c_1 + c_2 = 3,\ c_1 - c_2 = 5 \to c_1 = 4,\ c_2 = -1$ — yes.

4.5.8 Column space $C(A)$

$$A = [a_1 a_2 ... a_n] \to Ax = x_1 a_1 + ... + x_n a_n;$$
$$C(A) = Col(A) = \text{span} a_1, ..., a_n$$

The product $Axisa$ linear combination of columns of A. Column space $C(A) = \text{Col}(A)$ — the linear span of columns, a subspace of $R^m foranm \times n$ matrix A. Example: $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 0 \end{bmatrix}$ — columns are proportional, $\text{Col}(A) = \text{span}\{(1,\ 2,\ 0)\}$ — a line in R^3.

$$system Ax = b is solvable \Leftrightarrow b \in C(A)$$

The system $Ax = bhasa$ solution if and only if b lies in the column space: the solution x picks the coefficients of the combination of columns that gives b. The main criterion for solvability is weak.

4.5.9 Null space $N(A)$

$$N(A) = Null(A) = x : Ax = 0$$

Null space — the set of solutions to the homogeneous system; it is a subspace of R^n: if $Au = Av = 0$, then $A(su + tv) = 0$. Example: $A = \begin{bmatrix} 1 & 2 \end{bmatrix}$: $x_1 + 2x_2 = 0$ gives $x_1 = -2x_2$, therefore $\text{Null}(A) = \text{span}\{(-2,\ 1)\}$ — a line.

$$x : Ax = 0 — always subspace; x : Ax = b, b \neq 0 — not subspace (doesn't contain 0)$$

Solutions to the homogeneous system form a subspace; solutions to the non-homogeneous ($b \neq 0$) — no: zero vector does not belong to them. If the system $Ax = b$ is solvable, its solutions are a shift of the $null$ space: $x = x_p + x_n$, where $x_p isa$ particular solution, $x_n \in$ Null(A) (affine set). By week 8 midterm: be able to compute $A^T and (AB)^T$, use $P^{-1} = P^T and PA = LU$, test closure, find spans and membership in $C(A)$.

[Source: Week 7 Class problems (Solved), Linear Algebra I (Fall 2026) — 20 in-class problems; arithmetic verified in python]

Problem 1. Transpose of 2×2. Compute $A^T for A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

Solution. Transpose swaps rows and columns: $(A^T)_{ij} = a_{ji}$. Therefore $A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$; in particular, $(A^T)_{12} = a_{21} = 3$, diagonal remains unchanged.

Problem 2. Transposition of a sum. Compute $(A+B)^T for A = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}$.

Solution. First, add: $A + B = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$. Then transpose: $(A + B)^T = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$. Check: $A^T + B^T = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$ — matches.

Problem 3. Product $A^T A$. Find $A^T A for A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$.

Solution. $A^T A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$ — the result is symmetric, as always (exercise $26 ISA$).

Problem 4. Transposition of a product. Find $(AB)^T for A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}, B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$.

Solution. $AB = \begin{bmatrix} 3 & 3 \\ 2 & 6 \end{bmatrix}$, so $(AB)^T = \begin{bmatrix} 3 & 2 \\ 3 & 6 \end{bmatrix}$. The identity gives the same: $B^T A^T = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 3 & 6 \end{bmatrix}$ — the order of factors is reversed.

Problem 5. Checking symmetry. Is the matrix $A = \begin{bmatrix} 2 & -3 \\ -3 & 1 \end{bmatrix}$ symmetric?

Solution. $A^T = \begin{bmatrix} 2 & -3 \\ -3 & 1 \end{bmatrix} = A$. Yes, it is symmetric: the corresponding off-diagonal elements coincide ($a_{12} = a_{21} = -3$), and the diagonal does not change under transposition.

Problem 6. Symmetric and skew-symmetric parts. Decompose $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ into symmetric and skew-symmetric parts.

Solution. $A^T = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$; $S = (A + A^T)/2 = \begin{bmatrix} 1 & 3 \\ 3 & 3 \end{bmatrix}$ ($off-$diagonal elements are averaged), $K = (A - A^T)/2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Check: $S + K = A, S^T = S, K^T = -K$.

Problem 7. Row permutation: PA. Find PA, where $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

Solution. $PA = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$: the first $rowof$ the product is the second $rowofA$, and vice versa. Left multiplication swaps rows $1and2$.

Problem 8. Column permutation: AP. Find $APfor$ the same $AandP(A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix})$.

Solution. $AP = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$: the first column of the product is the second column of A, and vice versa. Right multiplication swaps columns $1and2$; note that $AP \neq PA$.

Problem 9. Inverse of a permutation matrix. Find $P^{-1}forP = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$.

Solution. For the permutation matrix $P^{-1} = P^T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$. Verification: $P^T P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = I$ — the columns of P are orthonormal.

Problem 10. Exchange rows $1and3$. Find the permutation matrix P that exchanges rows $1and3ofa3 \times 3$ matrix.

Solution. Exchange of rows $1and3$, applied to I_3: $P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$. For any A, the product PA exchanges its first and third rows; for example, for $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ $weget$ $\begin{bmatrix} 7 & 8 & 9 \\ 4 & 5 & 6 \\ 1 & 2 & 3 \end{bmatrix}$.

Problem 11. $PA = LU$ with zero pivot. Find P, L, $UforA = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$.

Solution. The pivot element $a_{11} = 0$ — exchange is unavoidable. Exchange the rows: $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $PA = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$. Below the pivot $3is0$ — elimination is complete, no multipliers: $L = I_2$, $U = PA = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$; result: $PA = LU$.

Problem 12. LU **without** row **exchanges. Find** L, $UforA = \begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix}$ **without** row **exchanges.**

Solution. The pivot element $2 \neq 0$, no exchanges needed; the multiplier $m_{21} = \frac{6}{2} = 3$. Step $R2 \leftarrow R2 - 3R1$: $\begin{bmatrix} 6 & 5 \end{bmatrix} - 3 \cdot \begin{bmatrix} 2 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \end{bmatrix}$, so $U = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$, $L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$. Verification: $LU = \begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix} = A$.

Problem 13. Subspace: line through origin. $IsW =$ **{**(x, y)**:** $y =$ **3x}** a **subspace of** R^2**?**

Solution. Yes. $W = \{t(1, 3)\colon\ t \in R\}$: the zero vector corresponds to $t = 0$. For $u = (s,$ 3s$), v = (t,$ 3t$) and any a, b$: au $+bv = (as+bt, 3(as+bt)) \in W$ — closure holds; $Wisa$ subspace (line through origin).

Problem 14. Counterexample: shifted line. $IsS =$ **{**(x, y)**:** $y =$ **3x** $+2$**}** a **subspace?**

Solution. No. The zero vector does not satisfy the equation: $3 \cdot 0 + 2 = 2 \neq 0$. The zero vector test is sufficient for refutation: a subspace must contain the origin.

Problem 15. Subspace: plane. $IsW =$ **{**(x, y, z)**:** $x + y - z = 0$**}** a **subspace?**

Solution. Yes. For $f(x,\ y,\ z) = x + y - z and u,\ v \in W\colon\ f(au + bv) = a \cdot f(u) + b \cdot f(v) = 0$, so au $+bv \in W$; and $f(0, 0, 0) = 0$. $Wisa$ subspace (plane through origin).

Problem 16. Linear combination of vectors. Express $(4, 1) asa$ **combination of vectors** $(1, 1) and (1, -1)$**.**

Solution. We seek $c_1(1,\ 1) + c_2(1,\ -1) = (4,\ 1)\colon\ c_1 + c_2 = 4,\ c_1 - c_2 = 1$. Adding: $2c_1 = 5$, $c_1 = \frac{5}{2}$, $c_2 = \frac{3}{2}$. Verification: $(\frac{5}{2} + \frac{3}{2},\ \frac{5}{2} - \frac{3}{2}) = (4,\ 1)$.

Problem 17. Span of proportional vectors. Find the linear span **of vectors** $(1,$ 2, 0$) and (2,$ 4, 0**).**

Solution. The second vector is twice the first: $(2, 4, 0) = 2(1, 2, 0)$, so any combination is a multiple of (1, 2, 0). Result: span{(1, 2, 0), (2, 4, 0)} $= \{t(1, 2, 0)\colon\ t \in R\}$ — a line in R^3.

Problem 18. Null space of $a 1 \times 3$ **matrix. Find Null**$(A) forA = \begin{bmatrix} 1 & 2 & -1 \end{bmatrix}$**.**

Solution. Equation $Ax = 0\colon\ x_1 + 2x_2 - x_3 = 0$. Free variables $x_2 = s, x_3 = t$, then $x_1 = -2$s $+ tandx = s(-2,\ 1,\ 0) + t(1,\ 0,\ 1)$. Result: Null$(A) =$ span{(−2, 1, 0), (1, 0, 1)} — a plane in R^3. Verification: $1 \cdot (-2) + 2 \cdot 1 - 0 = 0 and 1 \cdot 1 + 2 \cdot 0 - 1 = 0$.

Problem 19. Column space. Find Col$(A) forA = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 0 \end{bmatrix}$**; does the vector (3, 6, 0) belong to it?**

Solution. Columns $a_1 = (1,\ 2,\ 0) and a_2 = (2,\ 4,\ 0) = 2a_1$ are proportional, so $\mathrm{Col}(A) = \mathrm{span}\{(1,\ 2,\ 0)\}$ — a line in R^3. Since $(3, 6, 0) = 3(1, 2, 0)$, the vector lies on this line: yes (for example, $A \cdot \begin{bmatrix} 3 \\ 0 \end{bmatrix} = (3,\ 6,\ 0)$).

Problem 20. Transpose and determinant of 3×3. For $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ find $A^T and \det(A)$.

Solution. $A^T = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$. A is upper triangular, so $\det A = 1 \cdot 1 \cdot 2 = 2$ (product of diagonal elements); transposition doesn't change the determinant: $\det(A^T) = 2$.

[Source: Week 7 Independent Study Assignments, Linear Algebra I (Fall 2026) — 50 problems without solutions; solutions created

and verified in python]

Independent Study Assignments — ungraded self-practice (instructions: column vectors, intermediate steps, justification of subspace answers). File parts: $A(1-12)$ — transpositions, $B(13-20)$ — permutations and pivoting, $C(21-32)$ — subspace criteria, $D(33-42)$ — linear spans and matrix subspaces, $E(43-46)$ — mixed applications, $F(47-50)$ — advanced problems. Exercises 1–20 exactly match classroom problems 1–20 (below — compact solutions, complete ones above); the remaining 30 are fully solved.

Exercise 1. Compute $A^T for A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

Solution. $A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$ (rows became columns); in detail — problem 1 above.

Exercise 2. Compute $(A+B)^T for A = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}$.

Solution. $A + B = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$, so $(A+B)^T = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix} = A^T + B^T$.

Exercise 3. Find $A^T A for A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$.

Solution. $A^T A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$ — the result is symmetric.

Exercise 4. Find $(AB)^T for A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$.

Solution. $AB = \begin{bmatrix} 3 & 3 \\ 2 & 6 \end{bmatrix}$, so $(AB)^T = \begin{bmatrix} 3 & 2 \\ 3 & 6 \end{bmatrix} = B^T A^T$.

Exercise 5. Is the matrix $A = \begin{bmatrix} 2 & -3 \\ -3 & 1 \end{bmatrix}$ symmetric?

Solution. Yes: $A^T = \begin{bmatrix} 2 & -3 \\ -3 & 1 \end{bmatrix} = A$ — the off-diagonal elements coincide ($a_{12} = a_{21} = -3$).

Exercise 6. Decompose $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ into symmetric and skew-symmetric parts.

Solution. $S = (A + A^T)/2 = \begin{bmatrix} 1 & 3 \\ 3 & 3 \end{bmatrix}$, $K = (A - A^T)/2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$; check: $S + K = A$.

Exercise 7. Find PA, where $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

Solution. $PA = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$ — left multiplication swapped rows $1 and 2$.

Exercise 8. Find $AP for$ the same $A and P (A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix})$.

Solution. $AP = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$ — right multiplication swapped columns $1 and 2$; note that $AP \neq PA$.

Exercise 9. Find $P^{-1} for P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$.

Solution. $P^{-1} = P^T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$; check: $P^T P = I$.

Exercise 10. Find the permutation matrix P that swaps rows $1 and 3 of a 3 \times 3$ matrix.

Solution. $P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ — swapping rows $1 and 3 in I_3$; PA swaps the first and third rows of any matrix.

Exercise 11. Find P, L, $U for A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$.

Solution. Zero pivot requires exchange: $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $PA = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$; elimination is already complete, so $L = I_2, U = PA$; result: $PA = LU$.

Exercise 12. Find L, $U for A = \begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix}$ without row exchanges.

Solution. $m_{21} = \frac{6}{2} = 3$; step $R2 \leftarrow R2 - 3R1$ gives $U = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$; $L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$; check: $LU = A$.

Exercise 13. $IsW = \{(x, y)\colon\ y = 3x\}$ a **subspace of** R^2**?**

Solution. Yes: $W = \{t(1, 3)\colon\ t \in R\}$, zero vector when $t = 0$, and $a(s,\ 3s) + b(t,\ 3t) = (as + bt, 3(as + bt)) \in W$.

Exercise 14. $IsS = \{(x, y)\colon\ y = 3x + 2\}$ a **subspace?**

Solution. No: (0, 0) does not satisfy the equation $(0 \neq 3 \cdot 0 + 2)$ — no zero vector.

Exercise 15. $IsW = \{(x, y, z)\colon\ x + y - z = 0\}$ a **subspace?**

Solution. Yes: $f(x, y, z) = x + y - z$ is linear, $f(0,\ 0,\ 0) = 0 and f(au + bv) = a \cdot f(u) + b \cdot f(v) = 0 for u,\ v \in W$.

Exercise 16. Express $(4, 1) as a$ **combination of vectors** $(1, 1) and (1, -1)$**.**

Solution. $c_1 + c_2 = 4$, $c_1 - c_2 = 1 \to c_1 = \frac{5}{2}$, $c_2 = \frac{3}{2}$; verification: $(\frac{5}{2})(1,\ 1) + (\frac{3}{2})(1,\ -1) = (4,\ 1)$.

Exercise 17. Find the linear span **of vectors** $(1,\ 2,\ 0) and (2,\ 4,\ 0)$**.**

Solution. $(2, 4, 0) = 2(1, 2, 0)$, so span{(1, 2, 0), (2, 4, 0)} $= \{t(1, 2, 0)\colon\ t \in R\}$ — a line.

Exercise 18. Find Null$(A) for A = \begin{bmatrix} 1 & 2 & -1 \end{bmatrix}$**.**

Solution. $x_1 + 2x_2 - x_3 = 0 \to x_1 = -2s + t, x_2 = s, x_3 = t$; Null$(A)$ = span{(−2, 1, 0), (1, 0, 1)}.

Exercise 19. Find Col$(A) for A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 0 \end{bmatrix}$**; does (3, 6, 0) belong to it?**

Solution. Columns are proportional: Col(A) = span{(1, 2, 0)}; since $(3, 6, 0) = 3(1, 2, 0)$, the vector belongs to Col(A).

Exercise 20. For $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ **find** $A^T and \det(A)$**.**

Solution. $A^T = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$; A is upper triangular, so $\det A = 1 \cdot 1 \cdot 2 = 2 = \det(A^T)$.

Exercise 21. Dimensions. For $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ **compute** A^T **and specify dimensions.**

Solution. $A is 2 \times 3$; $A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} is 3 \times 2$: dimensions swap, $(m \times n)^T = n \times m$.

Exercise 22. Proof (sum)**. Prove** $(A + B)^T = A^T + B^T$ **element-wise.**

Solution. For $A = (a_{ij}), B = (b_{ij})$ of size $m \times n$ and each i, j: $((A + B)^T)_{ij} = (A + B)_{ji} = a_{ji} + b_{ji} = (A^T)_{ij} + (B^T)_{ij} = (A^T + B^T)_{ij}$. Elements match — matrices are equal.

Exercise 23. Proof (product). Prove $(AB)^T = B^T A^T$ **for compatible matrices.**

Solution. Let $Abem \times n, Bben \times p$. For each i, j: $((AB)^T)_{ij} = (AB)_{ji} = \Sigma_k a_{jk} b_{ki} = \Sigma_k (B^T)_{ik} (A^T)_{kj} = (B^T A^T)_{ij}$. Thus, $(AB)^T = B^T A^T$: the order of factors is reversed.

Exercise 24. Product $A^T Afor3 \times 2$. Compute $A^T AforA = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 3 \end{bmatrix}$ ($inthe$ file, the 3×2 matrix is written row-wise).

Solution. $A^T A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 2 & 10 \end{bmatrix}$ — the result is symmetric, as theory predicts (exercise 26).

Exercise 25. Symmetric and skew-symmetric parts. For $A = \begin{bmatrix} 1 & 3 \\ -2 & 4 \end{bmatrix}$ find the symmetric and skew-symmetric parts.

Solution. $A^T = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$. $S = (A + A^T)/2 = [1\frac{1}{2};\ \frac{1}{2}4]$, $K = (A - A^T)/2 = [0\frac{5}{2};$ −5/2 0]. Verification: $S + K = A, S^T = S, K^T = -K$.

Exercise 26. Symmetry of $A^T A$. Show that $A^T A$ is symmetric for any real matrix A.

Solution. $(A^T A)^T = A^T (A^T)^T = A^T A$ (rule $(AB)^T = B^T A^T$): the matrix equals its transpose, $i.e.$, is symmetric (for A of size $m \times n$, the product $A^T A$ is square $n \times n$).

Exercise 27. Scalar and outer product. Compute $u^T vanduv^T foru = (2, -1)^T$, $v = (3, 4)^T$.

Solution. $u^T v = 2 \cdot 3 + (-1) \cdot 4 = 2$ — a number (scalar product); uv^T $= \begin{bmatrix} 2 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 3 & 4 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ -3 & -4 \end{bmatrix}$ — $a2 \times 2$ matrix (outer product).

Exercise 28. Permutation of rows $1and2$. Construct the permutation matrix that exchanges rows $1and2of I_3$.

Solution. $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ — I_3 with rows $1and2$ exchanged. For any A with three rows, PA swaps the first two: $P \cdot \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix}$.

Exercise 29. $PAvsAP$. For $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ compare $PAandAPforA = \begin{bmatrix} 2 & 5 \\ 7 & 1 \end{bmatrix}$.

Solution. $PA = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 2 & 5 \\ 7 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 1 \\ 2 & 5 \end{bmatrix}$ — rows are permuted; $AP = \begin{bmatrix} 2 & 5 \\ 7 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 1 & 7 \end{bmatrix}$ — columns are permuted. $PA \neq AP$: left multiplication acts on rows, right on

columns.

Exercise 30. Verification $P^TP = I$. For $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ verify $P^TP = I$.

Solution. $P^T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$; $P^TP = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = I$. Rows of P^T are columns of $P(e_2, e_3, e_1)$, their scalar products give δ_{ij}; consequence: $P^{-1} = P^T$.

Exercise 31. Determinant of permutation. Find the determinant of the permutation matrix that exchanges rows $1 and 3$.

Solution. $P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$; one row exchange changes the sign: $\det P = -1$. Verification by expansion along the first row: $1 \cdot (-1)^{1+3} \cdot \det \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = 1 \cdot 1 \cdot (-1) = -1$.

Exercise 32. $PA = LU for 2 \times 2$. Find P, L, $U for A = \begin{bmatrix} 0 & 1 \\ 4 & 2 \end{bmatrix}$.

Solution. The pivot $a_{11} = 0$ — exchange is needed: $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $PA = \begin{bmatrix} 4 & 2 \\ 0 & 1 \end{bmatrix}$. Below the pivot $4 is 0$, elimination is complete: $L = I_2$, $U = \begin{bmatrix} 4 & 2 \\ 0 & 1 \end{bmatrix}$; result: $PA = LU$.

Exercise 33. Partial pivoting. Apply partial pivoting to $A = \begin{bmatrix} 0.001 & 1 \\ 2 & 1 \end{bmatrix}$; compare the multipliers before and after the exchange.

Solution. Without pivoting: pivot element 0.001, multiplier $m_{21} = \frac{2}{0}.001 = 2000$ — huge (subtracting $2000 \cdot$(first row) would strongly amplify errors). Partial pivoting takes the largest by magnitude candidate in column 1 (number 2); after exchange ($P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $PA = \begin{bmatrix} 2 & 1 \\ 0.001 & 1 \end{bmatrix}$) the multiplier $m_{21} = 0.\frac{001}{2} = 0.0005$. The exchange reduced the multiplier from $2000 to 0.0005 (bya$ factor of $4 \cdot 10^6$).

Exercise 34. Subspace? $x + 2y = 0$. Check if $W = \{(x, y): x + 2y = 0\}$ is a subspace.

Solution. Yes. $(1) 0 + 2 \cdot 0 = 0$ — the zero vector is in W. (2) From $x + 2y = 0 and x' + 2y' = 0$ follows $(x + x') + 2(y + y') = 0$. $(3)(cx) + 2(cy) = c(x + 2y) = 0$. Conclusion: $W = \text{span}\{(-2, 1)\}$ — a line through the origin.

Exercise 35. Subspace? $x + 2y = 3$. Check if $W = \{(x, y): x + 2y = 3\}$ is a subspace.

Solution. No: $0 + 2 \cdot 0 = 0 \neq 3$ — there is no zero vector in W, but a subspace must contain $0 (the$ shifted line does not pass through the origin).

Exercise 36. Subspace? $x \geq 0$. Check if $W = \{(x, y): x \geq 0\}$ is a subspace.

Solution. No. The zero vector is in W, and W is closed under addition (sum of non-negative x-coordinates is non-negative). But closure under scalar multiplication is violated: $(1,0) \in W$, but $(-1)\cdot(1,0) = (-1,0) \notin W$. A half-plane is not a subspace.

Exercise 37. Subspace? $x - y + z = 0$. Check if $W =$ {(x,y,z): $x - y + z = 0$} is a subspace.

Solution. Yes: $f(x,y,z) = x - y + z$ is linear, $f(0,\,0,\,0) = 0 and f(au + bv) = a \cdot f(u) + b \cdot f(v) = 0 for u,\ v \in W$. Explicitly: $x = y - z$ gives $(x,\ y,\ z) = y(1,\ 1,\ 0) + z(-1,\ 0,\ 1)$, that is $W =$ span{(1, 1, 0), (−1, 0, 1)} — a plane through the origin.

Exercise 38. Linear combination. Express $(6,2) as a$ combination of vectors $(1,1) and (1,-1)$.

Solution. $c_1 + c_2 = 6$, $c_1 - c_2 = 2 \to c_1 = 4$, $c_2 = 2$. Check: $4(1,1) + 2(1,-1) = (6,2)$.

Exercise 39. Geometry of span**. Describe geometrically** span**{(1, 2, 0), (0, 1, 2)}.**

Solution. The vectors are not proportional — the spanis two-dimensional: a plane through the origin. Its points: $s(1,2,0) + t(0,1,2) = (s,$ 2s $+t,$ 2t); eliminating parameters $(s = x, t = z/2) we$ get 4x − 2y $+z = 0$. Check: $4 \cdot 1 - 2 \cdot 2 + 0 = 0 and 4 \cdot 0 - 2 \cdot 1 + 2 = 0$.

Exercise 40. Null space of 2×3. Find Null$(A) for A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ (2×3 matrix, written row-wise).

Solution. From $row 2$: $x_2 = -x_3$; from $row 1$: $x_1 = -x_2 - x_3 = 0$. Free variable $x_3 = t$: $x = t(0,-1,1)$. Conclusion: Null(A) = span{(0, −1, 1)} — a line in R^3. Check: $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

Exercise 41. Column space of 3×2. Find Col$(A) for A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$.

Solution. Columns $a_1 = (1,\ 2,\ 3) and a_2 = (2,\ 4,\ 6) = 2a_1$ are proportional, so any combination is a multiple of a_1: Col(A) = span{(1, 2, 3)} — a line in R^3 (vectors $(t,$ 2t, 3t)).

Exercise 42. Membership in column space. Does (1, 1, 1) belong to the column space Col$(A) for A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$ (3×2 matrix)?

Solution. We seek $c_1(1,\ 0,\ 1) + c_2(0,\ 1,\ 1) = (1,\ 1,\ 1)$: the first two coordinates give $c_1 = c_2 = 1$, but then the third is $2 \neq 1$ — contradiction. Conclusion: $(1,1,1) \notin$ Col(A) (Col(A) is the plane $z = x + y$, and $1 \neq 1 + 1$).

Exercise 43. Intersection of subspaces. Prove that the intersection of two subspaces is a subspace.

Solution. For subspaces $U, W \subseteq V$: $(1) 0 \in U \cap W$. $(2) If u, v \in U \cap W$, then $u + v \in U and u + v \in W$ (closure of each), so $u + v \in U \cap W$. $(3) If u \in U \cap W$, then cu $\in U and cu \in W$, so cu $\in U \cap W$. The test is passed.

Exercise 44. Union of subspaces. Give two subspaces whose union is not a subspace.

Solution. InR^2 take the axes $U = \{(x, 0)\}$ and $W = \{(0, y)\}$ — both are subspaces. In the union lie $(1, 0)and(0, 1)$, but their sum (1, 1) lies on neither axis. The union is not closed under addition — not a subspace.

Exercise 45. Symmetric 3×3 matrices. $Istheset$ of symmetric 3×3 matrices a subspace of the space of all real 3×3 matrices?

Solution. Yes. (1) The zero matrix is symmetric. $(2)(S + T)^T = S^T + T^T = S + T$ for symmetric S, T. $(3)(cS)^T = c \cdot S^T =$ cS. Closure is satisfied — it is a subspace.

Exercise 46. Polynomials with $p(0) = 1$. $Istheset$ of polynomials satisfying $p(0) = 1a$ subspace?

Solution. No. The zero polynomial gives $p(0) = 0 \neq 1$ — there is no zero vector. Additionally, for $p(x) = 1andq(x) = 1 + x$: $(p + q)(0) = 2 \neq 1$ — closure under addition is also violated. (Polynomials with $p(0) = 0$, on the other hand, do form a subspace.)

Exercise 47. Advanced problem. Prove that Null$(A^TA) =$ Null$(A)for$ any real matrix A.

Solution. Two inclusions. $(1)IfAx = 0$, then $A^TAx = A^T(Ax) = 0$, that is, Null$(A) \subseteq$ Null(A^TA). (2) Let $A^TAx = 0$; multiply on the left by x^T: $x^TA^TAx = ||Ax||^2 = 0$. The sum of squares of the coordinates of Ax is zero, which means $Ax = 0$. Conclusion: Null$(A^TA) =$ Null(A); a consequence is that $AandA^TA$ have the same $null$ space and rank (least squares fact).

Exercise 48. Advanced problem. $IfPisa$ permutation matrix, prove that Col$(PA) = P$ Col(A).

Solution. By definition, Col$(PA) =$ {PAx: $x \in R^n$ }, P Col$(A) =$ {Pb: $b \in$ Col(A)}. $(1)Ify =$ PAx, then $y = P(Ax)$ with $Ax \in$ Col(A) — thus $y \in P$ Col(A). $(2)Ify = Pb$ with $b = Ax \in$ Col(A), then $y =$ PAx $\in$ Col(PA). Conclusion: Col$(PA) = P$ Col(A) — permuting rows permutes each point of the column space by the same permutation $(b \to Pbis$ bijective, P is invertible).

Exercise 49. Advanced problem. $IfWisa$ subspace, prove that span$(W) = W$.

Solution. span$(W)istheseto f$ all finite combinations $c_1w_1 + ... + c_kw_k$ of elements $w_i \in W$. (1) Each combination lies in W: $c_iw_i \in W$ (scalars), sum of elements of $WisinW$ (addition). (2) Each $w \in Wis$ the combination $1 \cdot w$. Conclusion: span$(W) = W$ — a subspace already contains all its linear combinations.

Exercise 50. Advanced problem. For $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ describe Col(A) and Null(A).

Solution. Columns $a_1 = (1, 0, 0)$, $a_2 = (2, 1, 0)$, $a_3 = (0, 1, 0) = a_2 - 2a_1$; the third row is zero, so all columns lie in the plane $z = 0$, and a_1, a_2 are independent. Result: Col$(A) =$ span{(1, 0, 0), (2, 1, 0)} = $\{(s + 2t, t, 0)\}$ — the entire plane $z = 0$. Null space: from $row2x_2 = -x_3$, from $row1x_1 = -2x_2 = 2x_3$; $x_3 = t$: $x = t(2, -1, 1)$. Result: Null$(A) =$ span{(2, −1, 1)} — a line. Verification: $A \cdot \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.